

Generally Recognized as Safe (GRAS) Notification

***Alternansucrase from *Leuconostoc mesenteroides* produced by
Escherichia coli EBASSC***

Submitted
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List of Abbreviations

%	Percentage
µg	Microgram
µmol	Micromole
ADI	Acceptable Daily Intake
AOAC	Association of Official Agricultural Chemists
ATCC	American Type Culture Collection
ATE	Acute Toxicity Estimate
bw	Body Weight
°C	Degrees Celsius
CAS	Chemical Abstract Registry Number
CFR	Code of Federal Regulations
CFU	Colony Forming Unit
cGMP	Current Good Manufacturing Practice
CLSI	Clinical and Laboratory Standards Institute
DNA	Deoxyribonucleic Acid
DP	Degrees of Polymerization
<i>E. coli</i>	<i>Escherichia coli</i>
E.C. number	Enzyme Commission Number
EDI	Estimated Daily Intake
EFSA	European Food Safety Authority
EHC	Environmental Health Criteria
EPA	Environment Protection Agency
FAO	Food and Agriculture Organization
FARRP	Food Allergy Research and Resource Program
FASTA	FAST-All
FASTER	Food Allergen Safety, Treatment, Education & Research Act
FCC	Food Chemicals Codex
FDA	U.S. Food and Drug Administration
FF	Final food
g	Gram
GHS	Globally Harmonized System
GI	Gastrointestinal
GILSP	Good Industrial Large Scale Practice
GLP	Good Laboratory Practice
GRAS	Generally Recognized As Safe
GRN	GRAS Notice
HACCP	Hazard Analysis and Critical Control Points
HFCS	High Fructose Corn Syrup
HPLC	High Performance Liquid Chromatography
Hrs	Hours
IFBC	International Food Biotechnology Council
IL	Interleukin

IP	Indian Pharmacopoeia
ISO	International Organization for Standardization
IUBMB	International Union of Biochemistry and Molecular Biology
JECFA	Joint FAO/WHO Expert Committee on Food Additives
kDa	Kilodalton
kg	Kilogram
L	Litre
<i>L. mesenteroides</i>	<i>Leuconostoc mesenteroides</i>
LAB	Lactic Acid Bacteria
LD ₅₀	Median Lethal Dose
LOQ	Limit of Quantitation
mg	Milligram
ml	Millilitre
MoS	Margin of Safety
n	Number
Ng	Nanogram
NIH	National Institutes of Health
NLT	Not Less Than
NMT	Not More Than
NOAEL	No Observed Adverse Effect Level
OECD	Organization for Economic Co-operation and Development
ORF	Open Reading Frame
PCR	Polymerase Chain Reaction
PID	Percentage Identity
PPM	Parts per million
QA	Quality Assurance
QPS	Qualified Presumption of Safety
RM	Raw Material
RNA	Ribonucleic Acid
SDAP	Structural database of allergenic proteins
SDS-PAGE	Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis
SGF	Simulated Gastric Fluid
Spp.	Species Pluralis (multiple species)
TMDE	Theoretical Maximum Daily Exposure
TMDI	Theoretical Maximum Daily Intake
TOS	Total Organic Solid
TSCA	Toxic Substances Control Act
U/g	Unit /grams
U/mg	Unit/milligram
USA	United States of America
US EPA	United States Environmental Protection Agency
USP	United States Pharmacopeia
WCB	Working Cell Bank
WHO	World Health Organization

Part 1. 21 CFR 170.225: Signed Statements and Certifications

1.1 GRAS Notice Submission

Advanced Enzyme Technologies Ltd. submits this GRAS notice in accordance with 21 CFR part 170, subpart E.

1.2 Name and Address of Notifier

APPLICANT

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PERSON RESPONSIBLE FOR THE DOSSIER

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AGENT WHO IS AUTHORIZED TO ACT ON BEHALF OF THE NOTIFIER

Kevin O. Gillies
Kevin O. Gillies Consulting Services, LLC
kevin@kogilliesconsulting.com

1.3 Name of Notified Substance

The notified substance is 'Alternansucrase' from *Leuconostoc mesenteroides* produced by genetically modified *Escherichia coli* EBASSC. In this GRAS notice, alternansucrase from *Leuconostoc mesenteroides* produced by genetically modified *Escherichia coli* EBASSC is referred to by names such as 'alternansucrase', 'ALS', 'AlSu', 'alternansucrase from *E. coli* EBASSC'.

1.4 Intended Conditions of Use

Alternansucrase from *E. coli* EBASSC is intended to be used in production of maltose alternan oligosaccharides, bakery process (bread making) and fruit processing for juice production.

The enzyme alternansucrase (E.C. 2.4.1.140) is intended to be used as a processing aid during food manufacturing. The enzyme is present in the food in insignificant amounts and does not have a

technical effect in the final food. The enzyme preparation is used at the minimum levels necessary to achieve the desired effect and according to requirements under current Good Manufacturing Practices. The general population is the target population for consumption.

1.5 Statutory Basis for GRAS Status

Advanced Enzyme Technologies Ltd. has determined that the intended use of alternansucrase from *E. coli* EBASSC is GRAS through scientific procedures in accordance with 21 CFR §170.30(a) and (b).

1.6 Premarket Exempt Status

Since Advanced Enzyme Technologies Ltd. has determined that the intended use of alternansucrase from *E. coli* EBASSC is GRAS, the use of the notified substance is exempt from pre-market approval requirements of the Federal Food, Drug, and Cosmetic Act.

1.7 Data Availability

Advanced Enzyme Technologies Ltd. agrees to make the data and information that are the basis for the evaluation of GRAS status available to FDA upon request. Such data and information may be sent by Advanced Enzyme Technologies Ltd. to FDA either in electronic format or on paper or reviewed during customary business hours at the above-mentioned address.

1.8 FOIA Statement

None of the data and information in this GRAS notice is exempt from disclosure under the Freedom of Information Act, 5 U.S.C. §552.

1.9 Certification

To the best of my knowledge, this GRAS notice is a complete, representative, and balanced submission that includes unfavorable information, as well as favorable information, known to me and pertinent to the evaluation of the safety and GRAS status of the intended use of alternansucrase from *E. coli* EBASSC.

1.10 FSIS Statement

Not applicable.

1.11 Signature of Responsible Party or Agent

Kevin O. Gillies



Kevin O. Gillies Consulting Services, LLC

kevin@kogilliesconsulting.com

Part 2. 21 CFR 170.230: Identity, Method of Manufacture, Specifications, and Physical or Technical Effect

2.1 Identity/ Identification

The subject of this GRAS notification is an alternansucrase produced by submerged fermentation of a genetically modified *Escherichia coli* (strain EBASSC) microorganism expressing the gene *asr* encoding for an alternansucrase from *Leuconostoc mesenteroides*.

2.1.1 SCIENTIFIC NAME, TAXONOMY AND OTHER NAMES

Name of the enzyme protein:	Alternansucrase
Synonyms:	Sucrose-1,6(3)- α -glucan 6(3)- α -glucosyltransferase; Sucrose:1,6-, 1,3- α -D-glucan 3- α - and 6- α -D-glucosyltransferase.
Systemic name:	Sucrose:1,6(1,3)- α -D-glucan 6(3)- α -D-glucosyltransferase
EC (IUBMB) number:	2.4.1.140
CAS number:	100630-46-4
Molecular weight:	224.8 kDa
Specificity:	Catalyzes a chemical reaction that transfers an alpha-D-glucosyl residue from sucrose, alternately to the 6- and 3-positions of the non-reducing terminal residue of an alpha-D-glucan, thereby creating a glucan (alternan) with alternating alpha-1,6- and alpha-1,3-bonds.
Amino acid sequence:	The total nucleotide and amino acid sequences have been determined

The classification of the enzyme according to the IUBMB is as follows:

EC 2	Transferases
EC 2.4	Glycosyltransferases
EC 2.4.1	Hexosyltransferases
EC 2.4.1.140	Alternansucrase

2.1.2 ENZYME ACTIVITY

Alternansucrase (IUBMB EC 2.4.1.140) catalyzes a chemical reaction that transfers an alpha-D-glucosyl residue from sucrose, alternately to the 6- and 3-positions of the non-reducing terminal residue of an alpha-D-glucan, thereby creating a glucan (alternan) with alternating alpha-1,6- and alpha-1,3-bonds.

The enzyme assay method is a company specific validated method. The enzyme activity was measured by high performance liquid chromatography (HPLC) using sucrose as the substrate. One unit of enzyme activity is defined as the amount of enzyme that releases 1 μmol of fructose in one minute under the given assay conditions.

The alternansucrase from *E. coli* EBASSC, described in this dossier possesses protease side activity. Protease activity was detected in food enzyme alternansucrase at approximately 100 U/g.

2.1.2.1 Information on the pH and temperature optima of the food enzyme

The activity of the food enzyme, alternansucrase from *E. coli* EBASSC was measured under various pH and temperature conditions.

Alternansucrase:

Activity

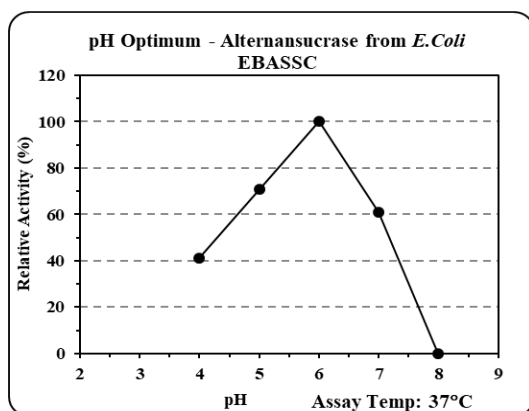


Figure 1: pH optimum of alternansucrase from *E. coli* EBASSC

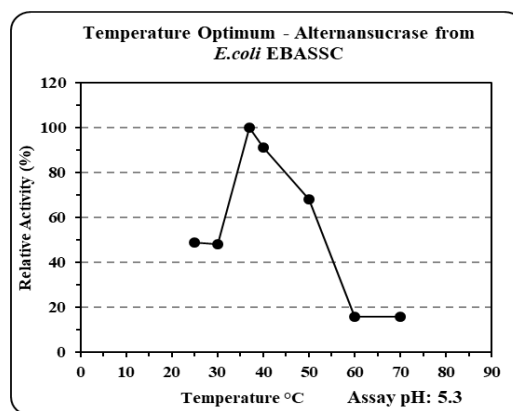


Figure 2: Temperature optimum of alternansucrase from *E. coli* EBASSC

The alternansucrase exhibited activity between pH 4.0 to 8.0 with optimum activity at pH 6.0 (Fig. 1) and at temperature ranging from 30°C to 60°C with optimum activity at 37°C (Fig. 2).

2.1.2.2 Information on the stability of the food enzyme

The stability of the food enzyme, alternansucrase from *E. coli* EBASSC was analyzed under various pH and temperature conditions.

Stability

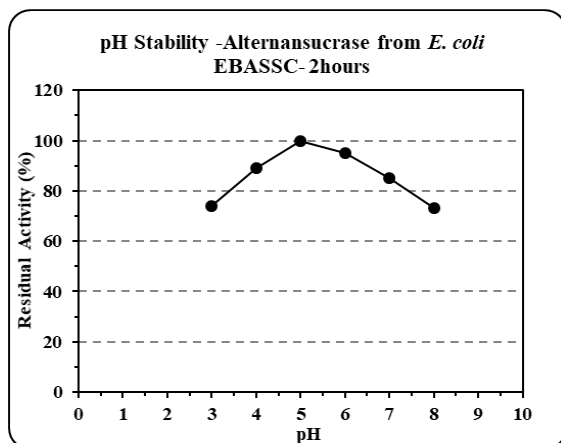


Figure 3: pH stability of alternansucrase from *E. coli* EBASSC

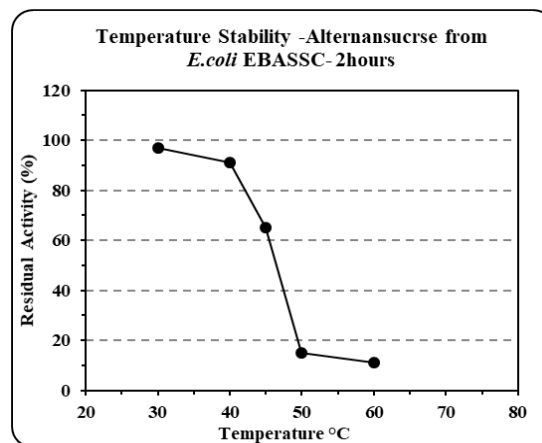


Figure 4: Temperature stability of alternansucrase from *E. coli* EBASSC

The alternansucrase from *E. coli* EBASSC was stable between pH 4.0 to 6.0 for 2 hours (Fig. 3). The enzyme was found to be stable between 30°C to 40°C for 2 hours. The enzyme activity decreases rapidly at temperatures higher than 40°C (Fig. 4).

2.2 Production Organism

2.2.1 SCIENTIFIC NAME, TAXONOMY AND OTHER NAMES

The microorganism used for the production of the alternansucrase is a genetically modified *Escherichia coli* strain EBASSC.

The strain is described as *E. coli* EBASSC throughout the dossier.

The production strain has been confirmed to be derived from *Escherichia coli* W3110 by the genetic modifications described in this notice by whole genome sequencing and phylogenetic analysis.

Taxonomy:

Name	: <i>Escherichia coli</i> EBASSC
Kingdom	: Bacteria
Phylum	: Proteobacteria
Class	: Gamma Proteobacteria
Order	: Enterobacteriales
Family	: Enterobacteriaceae
Genus	: <i>Escherichia</i>
Species	: <i>Coli</i>
Generic Name	: <i>Escherichia coli</i> EBASSC
Commercial Name	: <i>Escherichia coli</i> EBASSC

The name of the production organism in the dossier is *Escherichia coli* EBASSC. The strain designation *Escherichia coli* strain EBASSC, *Escherichia coli* (strain EBASSC), *Escherichia coli* EBASSC and *E. coli* EBASSC are used interchangeably while referring to the production strain, in certain sections of the dossier.

2.2.2 PRODUCTION STRAIN, RECIPIENT AND DONOR, STRAIN

Production strain:

The production strain *Escherichia coli* EBASSC is a non-pathogenic, nontoxigenic strain generated using recombinant DNA technology by modification of the host strain *E. coli* W3110. The strain is deposited with the American Type Culture Collection (ATCC), United States of America (USA) under strain designation SD-8485. The strain *E. coli* W3110 is a well characterized lab strain derived from *E. coli* K-12 (Bachman, 1972). Genomes of both strains have been sequenced and are publicly available (Hayashi et al., 2006). *E. coli* K-12 is the most widely used safe laboratory strain (Bauer et al., 2007). It is a non-pathogenic, non-toxigenic microorganism, with a history of safe use. Use of *E. coli* K-12 strain and its derivatives are exempt from the requirements of the NIH Guidelines (Meyer et al., 2019) indicating that the strain is not associated with disease in healthy adult humans. The genetically modified production organism complies with OECD (Organization for Economic Cooperation) and criteria for GILSP (Good Industrial Large Scale Practice) microorganisms and meets the criteria for a safe production microorganism as described by Pariza and Foster (Pariza & Foster,

1983) and later Pariza and Johnson (Pariza & Johnson, 2001) and several expert groups (IFBC, 1990; OECD, 1993; JECFA, 2006).

Recipient strain:

The recipient strain *E. coli* W3110, devoid of rhamnose utilization gene, was used in the construction of the alternansucrase production strain *E. coli* CAML357. The cassette used for generating the rhamnose mutant contained the chloramphenicol marker for selection which was eventually removed in the final host strain *E. coli* CAML357.

Donor strain:

The alternansucrase gene donor microorganism - *Leuconostoc mesenteroides* - is a member of lactic acid bacteria (LAB), which has a long history of safe use in food fermentations, industrial-scale enzyme production and consumption as a probiotic, can be considered as safe for production of enzymes for food as well as feed processing and numerous other industrial applications. Importantly, EFSA has assigned *L. mesenteroides* a “Status of Qualified Presumption of Safety” (QPS) (EFSA 2007, 2020) and is also included in the Inventory of Microbial Food Cultures with Safety Demonstration in Fermented Food Products by the International Dairy Federation (495/2018).¹ Thus, expressing alternansucrase gene *asr* from this organism, therefore, doesn’t cause a safety concern.

2.2.3 ALTERNANSUCRASE EXPRESSION PLASMID

The plasmid pAML369 contains the codon optimized sequence of the *asr* gene from *L. mesenteroides* under the control of the rhamnose inducible promoter in the recipient host strain *E. coli* CAML357. The production organism *E. coli* EBASSC is a host *E. coli* CAML357 transformed with a plasmid pAML369. Since the gene sequence is of synthetic origin, no part of the production strain or the expression plasmid contain original genetic elements from the donor strain. Production efficiency was increased by fusion of the periplasmic signal sequence to the *asr* open reading frame (ORF) to allow periplasmic secretion. Ectopic expression of the *asr* gene from the plasmid upon induction with rhamnose eventually leads to the production of alternansucrase enzyme.

2.2.4 ANTIBIOTIC GENES IN THE PRODUCTION STRAIN

In addition to the *asr* gene that encodes alternansucrase, the pAML369 also contains a kanamycin resistance gene as a selection marker. However, the enzyme product is free from the recombinant DNA including the *asr* gene and the kanamycin resistance gene.

Phenotypically, *E. coli* EBASSC was evaluated for its susceptibility to different antibiotics by following the Clinical and Laboratory Standards Institute’s guidelines (CLSI, 2015 and 2018) as recommended by European Food Safety Authority (2018). Three batches of *E. coli* EBASSC were

¹ Bulletin of the international dairy federation 495/2018

assessed for the susceptibility to eight antibiotics (e.g. ampicillin, gentamicin, kanamycin, streptomycin, tetracycline, ciprofloxacin, colistin and fosfomycin) using an agar dilution assay. The production strain was sensitive to seven antibiotics viz. ampicillin, gentamicin, streptomycin, tetracycline, fosfomycin, colistin and ciprofloxacin and resistant to kanamycin.

The resistance to kanamycin was due to the production strain *E. coli* EBASSC containing the plasmid harboring the *asr* gene which uses a kanamycin gene and resulting kanamycin resistance as a selection marker. However, the risk of transmission of the kanamycin gene is eliminated since the DNA from the production organism, including the kanamycin gene, is removed from the enzyme product during processing and was not detected in the final enzyme preparation (see Section 2.2.7).

2.2.5 INFORMATION ON GENETIC STABILITY OF PRODUCTION STRAIN

When implemented, the fermentation process starts from an identical replica of the production strain seed vial. Production stocks from the “Working Cell Bank” are used to start the fermentation process. A Working Cell Bank is a collection of vials containing a pure culture. The cell line history and the production of a WCB, propagation, preservation and storage are monitored and controlled. The WCB is prepared from a selected strain. A WCB vial is only accepted for production runs if its quality meets the required standards. This is determined by checking identity, viability, microbial purity and productivity of the WCB vial. The accepted WCB vial is used as seed material for the inoculum. Testimony to the stability of the strain is given by monitoring the growth behavior and by comparable levels of alternansucrase activity in a number of fermentation batches made with *E. coli* EBASSC. The activity measurements from parallel fermentations showed that the productivity of the production strain remains similar, indicating strain stability.

2.2.6 ABSENCE OF THE PRODUCTION ORGANISM IN THE PRODUCT

The absence of production organism (*E. coli* EBASSC) in the product was demonstrated following EFSA guidance (EFSA Journal 2021a).

A method was developed to check the presence of the production organism in the enzyme sample. The positive control, viz. the enzyme sample (1 g) spiked with 10 cells showed growth on the selected medium, indicating that the technique is suitable to detect the presence of production microorganism in the enzyme sample. The method was then used to analyze enzyme preparation product from three production batches for the presence or absence of the production microorganism *E. coli* EBASSC. The results indicated that the test samples did not form colonies of the recombinant production strain on petri plates. The lower detection limit for recombinant production strain is 10 cells per gram of sample.

2.2.7 ABSENCE OF TRANSFERABLE RECOMBINANT DNA SEQUENCES IN THE ENZYME PREPARATION

A PCR based method with detection limit of as low as 10 ng of DNA was developed to check the presence of transferrable recombinant DNA sequences from the production organism in the enzyme sample. A positive control, viz. the enzyme sample (1 g) spiked with 10 ng of total DNA from the

production organism *E. coli* EBASSC was used to validate the method of DNA recovery from the complex enzyme matrix. Three production batches of alternansucrase enzyme were analyzed for presence/absence of DNA from the production microorganism by using primers specific for the *asr* gene fragment and the kanamycin resistance gene. Amplification of these genes in the positive control indicated suitability of the method for detection of recombinant DNA sequences from enzyme sample. Absence of amplicon in the tested enzyme sample from production batches indicated that the enzyme samples were free of recombinant DNA [limit of detection 10 ng per gram of the enzyme sample].

2.2.8 **GOOD INDUSTRIAL LARGE-SCALE PRACTICE (GILSP)**

The production organism used for the production of alternansucrase covered in this dossier is genetically modified *E. coli* EBASSC.

E. coli strains such as *E. coli* K-12, B, C, W and their derivatives are all classified as risk group 1 organisms in biosafety guidelines. *E. coli* W3110 strain was derived from the well characterized wild-type parent *E. coli* K-12. These strains are frequently used for recombinant protein production and have a history of safe use. (Bauer et al., 2007, Gorbach, 1978)

In recent years, considerable knowledge and experience on the safe use of recombinant *E. coli* strains at industrial scale has been accumulated. *E. coli* K-12 strain can be used under the lowest containment level at large scale, GILSP, as defined by OECD (1993).

E. coli K-12 has been considered for over 30 years as safe to use for various biotechnological applications as it does not produce toxins harmful for ingestion. *E. coli* K-12 has been widely used in production of food ingredients, specialty chemicals and human drugs. Because of its history of safe use, it has been exempted from the Environment Protection Agency (EPA), USA review of Toxic Substances Control Act, 1967 (TSCA) (EPA, 1997). Common therapeutic proteins expressed in *E. coli* are Human insulin (Nilsson et al, 1996), interleukin 2 (IL-2) (Roifman, 1985), etc. Several industrial enzymes have also been produced in *E. coli* viz. xylanase (Whitehead and Hespell, 1989), betaglucosidase (Ferreira, 2018), and cellulases (Amraini et al, 2017).

The EFSA has provided positive scientific opinions for the use of substances produced by the selected strains of *E. coli* in food. (EFSA Journal 2014, EFSA Journal 2019)

The United States FDA has also issued “no questions” letters in response to GRAS notices for substances produced by an *E. coli* K-12 derivative and supporting its position as safe for use in food production (GRN 624). The production of rennin by *E. coli* K-12 was affirmed as GRAS by the FDA in 1990. Thus, *E. coli* strain K-12 and its derivatives such as W3110 can be considered as safe production organisms for enzymes used in food / feed processing as well as numerous other industrial applications.

2.2.9 **SECONDARY METABOLITES**

Most industrial strains of *E. coli* are from safe strain lineages that have been repeatedly tested according to the criteria established in the Pariza & Johnson publication (Pariza & Johnson, 2001). (see Section 6.5 for details)

The production strain *E. coli* EBASSC is derived from parental strain *E. coli* K-12, which is known for its safe usage. *E. coli* K-12 has been widely used as safe laboratory organism because it doesn't produce toxins that cause illness to human by ingestion (US EPA, 1997) such as shiga-like toxins produced by some other *E. coli* strains. *E. coli* K-12 lacks the ability to produce significant quantities of toxins that affect humans (US EPA, 1997)². No records of K-12 enterotoxin-induced disease for fermentation workers were reported so far (US EPA, 1997). Also *E. coli* K-12 doesn't carry known virulence genes. *E. coli* K-12 is observed to be non-competitive and poor colonizers, and this trait makes *E. coli* K-12 unlikely to inhabit the GI track of humans or other animals (Freter et al., 1983).

The production strain, *E. coli* EBASSC, does not produce secondary metabolites of toxicological concern to human consumers based upon its strain background and the characteristics of the plasmid pAML369. In addition, the strain, *E. coli* EBASSC, was evaluated for its antimicrobial activity following FAO/ WHO Volume 4 (2006) guidelines found not to produce antimicrobials. Heavy metal contents in the alternansucrase produced by *E. coli* EBASSC was within the specifications and below the detection limits of the test methods. ([Annex A](#)) The safety of the production organism was further substantiated by lack of toxicity when the alternansucrase from *E. coli* EBASSC was tested in standard *in vitro* and *in vivo* toxicity model systems (Refer to section 6.4 for details).

² US EPA (Environmental Protection Agency); Final risk assessment of *Escherichia coli* K-12 derivatives (1997)

2.3 Manufacturing process

2.3.1 OVERVIEW

Alternansucrase is produced by microbial fermentation process carried out following current Good Manufacturing Practice (cGMP) and the principles of Hazard Analysis and Critical Control Points (HACCP). The manufacturing facility is ISO 9001, ISO 22000 (Food Safety Management System) and GMP certified in accordance with 21CFR§110 and 21CFR§117.

Fermentation is a well-known process that occurs in food and has been used for the production of food enzymes for decades. Liquid state or submerged fermentation is used to produce the alternansucrase.

As shown below, the key steps for the production of the alternansucrase are fermentation, recovery, formulation, and packaging. The process is illustrated in Fig 5.

2.3.2 FERMENTATION

2.3.2.1 Raw materials

Materials used in the fermentation and post-fermentation processes (inoculum, seed, main fermentation, recovery, and formulation), are all food-grade GRAS or approved food additives in accordance with US federal regulations. The raw materials used in the fermentation conform to current Food Chemicals Codex specifications except for those that do not appear in the FCC. There are no ingredients based on milk, soy, or any of the named nine allergens in the Food Allergen Safety, Treatment, Education & Research Act (FASTER Act, 2021).

2.3.2.2 Inoculum (Seed)

A suspension of a pure culture of *E. coli* EBASSC cells is aseptically transferred to an inoculum flask containing fermentation medium.

Inoculum preparation is carried out by trained personnel in biosafety cabinets, installed in a class 100,000 cleanroom. Subsequent growth of the bacterium on a laboratory shaker is also carried out in a similar cleanroom environment. The purity of the initial inoculum is ensured by monitoring the inoculum's specific growth pattern, direct microscopic evaluation, gram staining, and reviewing the characteristics of viable growth on agar medium. The culture is grown in the flask under optimum conditions in order to obtain a sufficient amount of biomass, which is subsequently used as inoculum for the seed fermentation.

2.3.2.3 Seed Fermentation

The inoculum is aseptically transferred to the seed fermenter containing seed fermentation medium. When a sufficient amount of biomass is developed (typically up to 8 hours), the content of the seed fermenter is used for inoculation of the main fermentation.

2.3.2.4 Main fermentation

The fermentation process is conducted in an entirely closed and sterile environment. Enzyme production takes place at this stage. The fermentation is operated in fed batch mode. The production microorganism *E. coli* EBASSC grown in seed fermenter is aseptically transferred to the main fermenter containing the sterilized and cooled fermentation medium. After the desired growth is achieved in the fermenter, inducer is aseptically added in order to induce the synthesis of the alternansucrase enzyme, which is produced and retained intracellularly. The fermentation process is continued for a predetermined time or until laboratory test data show that the desired enzyme production has been obtained or that the rate of enzyme production has decreased below a predetermined production rate. When the desired enzyme level is reached, the fermentation is complete.

The samples are checked microscopically for contamination with other microorganisms during seed and fermentation stages at regular intervals. Microscopic observations for purity are also confirmed by plating the samples on nutrient agar and incubating plates at 37°C for 24 hrs. A production batch is stopped and discarded once any contamination is observed at any stage.

2.3.3 RECOVERY

The alternansucrase produced in the fermentation process is separated from the biomass and other insoluble at this stage.

During fermentation, the enzyme protein alternansucrase is produced and retained by the *E. coli* EBASSC, intracellularly. During recovery, the fermentation broth is subjected briefly to an elevated temperature with due consideration to enzyme stability and passed through a cell homogenizer to lyse the cells, this results in the release of the enzyme from the bacterial cells. The enzyme-containing fermentation medium is then separated from the biomass.

The steps of enzyme recovery include:

- Primary separation (biomass and insoluble/unutilized media from liquid)
- Centrifugation
- Concentration
- Centrifugation
- Pre-filtration and micro (germ) filtration
- Spray drying

2.3.3.1 Primary Separation

Filter aids are added to promote biomass flocculation and filtration is carried out at a controlled pH using a horizontal filter press. This enables separation of the biomass and other insoluble impurities from the soluble enzyme containing liquid. The primary separation is performed at defined pH and temperature ranges in order to minimize the loss of enzyme activity.

2.3.3.2 Centrifugation

The filtered liquid is passed through a high-speed centrifuge to separate and remove other insoluble and partial soluble impurities.

2.3.3.3 Concentration and centrifugation

The primary filtration step is followed by ultrafiltration and diafiltration of the filtrate from the primary recovery step to concentrate the enzyme containing filtrate and to reach the desired enzyme activity. Temperature and pH are controlled during this step. The filtered liquid is passed through a high-speed centrifuge to separate and remove other insoluble and partially soluble impurities.

2.3.3.4 Pre-filtration and micro-filtration (Germ filtration)

A filtration step on a dedicated micro (germ) filtration media is performed to ensure the removal of the production strain cells and insoluble substrate components from the fermentation. The concentrated enzyme liquid obtained after microfiltration, if required, is stabilized using suitable food grade GRAS or approved food additive stabilizer.

2.3.3.5 Spray Drying

The concentrated enzyme solution is spray-dried in presence of GRAS or approved food additive food-grade stabilizers (e.g., maltodextrin) to obtain the unformulated concentrate.

2.3.4 FORMULATION AND PACKAGING

Food enzyme alternansucrase is available in different enzyme activities.

If required, the spray-dried unformulated concentrate (not less than 3,000 U/g) is further formulated with approved food grade formulating agents and adjusted to a declared enzyme activity.

The alternansucrase is tested by Quality Control for the quality specifications and released by Quality Assurance. The final product is packed in suitable food packaging material before storage. Warehousing and transportation are performed according to specified conditions mentioned on the accordant product label for final preparations.

Allergenicity of enzyme preparations

The alternansucrase is not a known allergen (See section 6.2.2). Materials used in the fermentation process (inoculum, seed, and main fermentation) of alternansucrase are food-grade substances. There are no ingredients added based on milk, soy, or any of the listed nine allergens during fermentation process and post fermentation process causing allergenicity or labelling concerns. (FASTER Act, 2021). Allergen information for alternansucrase enzyme preparation is provided in [Annex B](#).

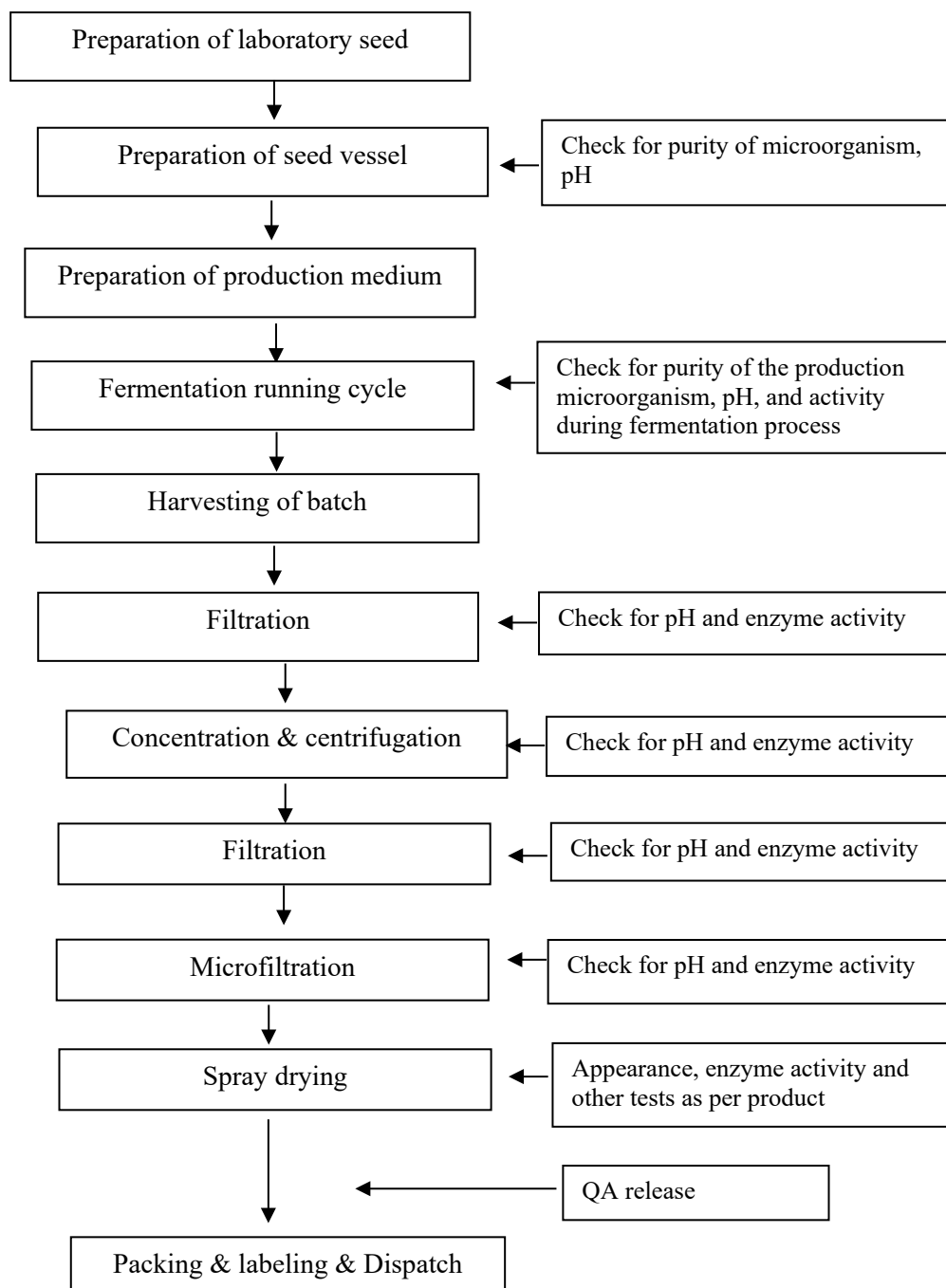


Figure.5: Manufacturing process of the alternansucrase from *E. coli* EBASSC

2.3.5 QUALITY CONTROL OF FINISHED PRODUCT

The proposed processing aid complies with the internationally accepted JECFA specifications for chemical and microbiological purity of food enzymes (FAO/WHO, 2006a, b), as mentioned in the table below.

Test	Limits	Reference
Heavy metals	Not more than 30 mg/kg	AOAC 2015.01, 21 st edition ³
Lead	Not more than 5 mg/kg	AOAC 2015.01, 21 st edition
<i>Salmonella</i> sp.	Absent in 25 g of sample	USP <62>
Total coliforms	Not more than 30 per gram	USP <2021>
<i>Escherichia coli</i>	Absent in 25 g of sample	USP <62>
Antimicrobial activity	Not detected	FAO/WHO, (2006b) Volume 4 ⁴

The specifications of the alternansucrase are described in Section 2.4.

The proof that the alternansucrase from *E. coli* EBASSC complies with these specifications is shown by the analysis on three batches. Refer to [Annex A](#) for analyses of heavy metals, and section 6.2.1 for antimicrobial activity.

³ AOAC Official Method 2015.01 Heavy Metals in Food. Inductively Coupled Plasma–Mass Spectrometry First Action 2015

⁴ FAO/WHO (2006b). Joint FAO/WHO Expert Committee on Food Additives Monographs: Combined compendium of food additive specifications, Volume 4, pp 112.

2.4 Product Specifications and Compositional Variability

2.4.1 *PRODUCT SPECIFICATIONS*

Specifications for alternansucrase preparation have been established by Advanced Enzyme Technologies Ltd. and are summarized in Table 1. All methods are standard methods validated for the purpose and references are provided in the below table.

Table 1: Product specifications for alternansucrase from *E. coli* EBASSC

Product specification	Advanced Enzyme Technologies Ltd.	
	Limits	Reference Method
Enzyme activity	Not less than 3,000 U/g	Internal method
Appearance/ Description	Light brown to brown colored powder with characteristics odor	Visual, olfactory
Moisture/ Loss on Drying	Not more than 7.0%	USP <731>
Heavy metal	Not more than 30.0 ppm	AOAC 2015.01, 21st Edition
Lead	Not more than 5.0 ppm	AOAC 2015.01, 21st Edition
Total coliform	Not more than 30 cfu/g	USP<2021>
<i>Escherichia coli</i>	Absent in 25g	USP <62>
<i>Salmonella</i> spp.	Absent in 25g	USP <62>
Total viable count	Not more than 10000 cfu/g	USP <62>
<i>Antimicrobial activity</i>	Absent by test	FAO/WHO, (2006b) Volume 4

2.4.2 COMPOSITIONAL VARIABILITY

Commercial enzymes, whether used in the production of food, feed, or for technological purposes, are biological isolates of variable composition. Apart from the enzyme protein in question, microbial food enzymes also contain some substances derived from the producing microorganism and the fermentation medium. These constituents consist of organic material and inorganic salts. As established by JECFA (FAO/WHO, 2006 (a),(b)), the percentages of these organic materials are summarized and expressed as Total Organic Solids (TOS)⁵.

2.4.2.1 Quantitative Composition

The relative purity of alternansucrase from *E. coli* EBASSC was measured, and the TOS values were calculated, in 3 batches after drying.

Table 2: Analysis of compositional variability of alternansucrase from *E. coli* EBASSC

Batch no	122101 ⁶	122103	122105	Mean enzyme activity
Ash (%)	10.26	11.37	10.85	-
Water (%)	4.56	6.20	5.29	-
TOS (%)	85.18	82.43	83.86	-
% Protein	79.15	71.77	79.13	-
Enzyme activity (U/g)	3960	3307	3955	3741
U/mg TOS ⁷	4.65	4.01	4.72	-
Unit Definition: One unit is defined as the amount of enzyme that releases 1 µmol of fructose in one minute under the given assay conditions.				

⁵ The TOS value is an internationally accepted method to describe the chemical composition of commercial food enzymes (TOS% = 100% – (Ash % + Water % + Diluent %)). The ratio between the enzyme activity and TOS is an indication of the relative purity of the enzyme.

⁶ The batch used for toxicity studies adheres to JECFA specifications for food enzyme(s).

⁷ U/mg TOS = Activity U/mg x 100 / TOS %

2.4.2.2 Data on batch-to-batch variability for relevant parameters

Three batches of alternansucrase from *E. coli* EBASSC were analyzed, and the results were compared with food-grade specifications. As shown in Table 3, all tested batches complied with the product specifications, demonstrating the suitability of the production process.

Table 3: Comparative analysis of three batches

Parameter	Specification	Batch		
		122101	122103	122105
Enzyme activity	Not less than 3000 U/g	3960 U/g	3307 U/g	3955 U/g
Description	Light brown to brown colored powder with characteristic odor	Light brown colored powder	Light brown colored powder	Light brown colored powder
Moisture/Loss on drying (%)	Not more than 7.0%	4.56%	6.20%	5.29%
Heavy Metal Analysis				
Heavy metals	Not more than 30 ppm	0.71 ppm	0.48 ppm	0.69 ppm
Lead	Not more than 5.0 ppm	0.1 ppm	0.1 ppm	0.1 ppm
Microbial Analysis				
Total viable count	Not more than 10000 cfu/g	220 cfu/g	1400 cfu/g	130 cfu/g
Total Coliform	Not more than 30 cfu/g	<10cfu/g	<10cfu/g	<10cfu/g
<i>E. coli</i>	Absent in 25g	Absent in 25g	Absent in 25g	Absent in 25g
<i>Salmonella</i> spp.	Absent in 25g	Absent in 25g	Absent in 25g	Absent in 25g
Antimicrobial activity	Absent by test	Absent by test	Absent by test	Absent by test

* LOQ for cadmium, lead and arsenic -0.1 ppm

* LOQ for mercury -0.025 ppm

2.5 Technical Effect

2.5.1 *MODE OF ACTION*

Alternansucrase catalyzes a chemical reaction that transfers an alpha-D-glucosyl residue from sucrose, alternately to the 6- and 3-positions of the non-reducing terminal residue of an alpha-D-glucan, thereby creating a glucan (alternan) with alternating alpha-1,6- and alpha-1,3-bonds.

2.5.2 *USES*

In principle, the enzymatic conversions of sucrose with the help of alternansucrase can be of benefit in the processing of all foods and food ingredients, which naturally contain sucrose.

Alternansucrase can be used for production of oligosaccharides by tailoring the reaction kinetics with the help of acceptor molecules. Alternansucrase acts on sucrose and produces long chain polymer called alternan consisting of alternating α -(1,6) and α -(1,3) linked D-glucosyl residues (Sanz et al., 2005; Wangpaiboon et al., 2018; Morales et al., 2001); and fructose is released as by-product. This enzyme uses different types of acceptor molecules to start polymerization reaction. The acceptor molecule drives the length and nature of oligosaccharide or polysaccharide formed. A well-known acceptor for alternansucrase is maltose (Bivolarski et al., 2021; Sanz et al., 2005; Cote et al., 2008); other saccharides such as raffinose (Cote et al., 2009), cellobiose (Morales et al., 2001), and leucrose (Cote et al., 2008) also serve as chain initiation sites for alternansucrase. The nature of alternan chains (degree of polymerization, composition, branching pattern etc.) formed in reaction solely depends on acceptor molecule and its ratio with respect to sucrose.

Compared to other polysaccharides, alternan has unique physical properties like low viscosity (than dextran) (Molina et al., 2020) and resistance to degradation by common digestive enzymes (Molina et al., 2020; Cote, 1992). Therefore, alternansucrase can be used in the production of dietary fibers/foods, which would have low glycemic index and prebiotic effects (Sanz et al., 2005).

Production of maltose alternan oligosaccharides:

Alternansucrase is intended to be used in the production of maltose alternan oligosaccharides using sucrose as a substrate and maltose as an acceptor molecule. Unlike other glucosyltransferases, alternansucrase utilizes a low-cost substrate (sucrose/ table sugar) for the production of oligosaccharides (Wangpaiboon et al., 2021). The enzyme forms α -(1,3) linkage with maltose only after the formation of α -(1,6) bond and incapable of forming two consecutive α -(1,3) linkages in a growing chain; the oligosaccharide formed does not contain more than two consecutive α -(1,6) linkages. It makes a polymer of a unique structure consisting alternating α -(1,3) and α -(1,6) bonds (Cote et al., 2008).

Alternansucrase produces panose as the first product after transferring glucose from sucrose to the maltose acceptor, which subsequently acts as an acceptor for enzyme. Formation of alternan oligosaccharides (i.e., DP lengths and their yields) is strongly influenced by maltose/sucrose ratio. At higher concentrations, maltose competes for glucose hydrolyzed from sucrose and leads to the

accumulation of oligosaccharides with lower DPs (Bivolarski et al., 2021). Therefore, the concentration of maltose in reaction mixture is the key factor for production of different types of maltose alternan oligosaccharides.

The production of alternan oligosaccharides achieved by continuous process where sucrose/refined sugar is added in a reactor at definite time intervals and by-product fructose is simultaneously removed from the system with the help of membrane filtration i.e., nanofiltration. The reaction is initiated by addition of maltose, sucrose and alternansucrase enzyme that are then incubated at 37°C up to 72 hrs (preferably 24 hrs). When the desired oligosaccharides are produced, the reaction is terminated followed by removal of high molecular weight alternans and alternansucrase enzyme using ultrafiltration. The maltose alternan oligosaccharide stream is then concentrated by water removal to obtain a stable syrup (water $\leq$ 28%) or spray dried to powder (water $\leq$ 8%).

The benefits of the conversion of sucrose with the help of alternansucrase in production of maltose alternan oligosaccharides are:

- Use of economical raw materials (commercial sucrose, maltose) to produce nutritionally important oligo/polysaccharides - Chemical synthesis of oligo/ polysaccharides is complex and expensive since it involves use of sugar derivatives as starting materials, which are less stable or require protection/ de-protection during synthesis. On the other hand, alternansucrase utilizes commonly available substrate like sucrose that makes production process viable at commercial scale.
- Mild reaction conditions - Alternansucrase requires mild reaction conditions such as temperature (37°C) and pH (4 to 6) for production of oligosaccharides unlike chemical synthesis methods, which require harsher reaction parameters.
- The reaction can be tailored (with acceptor molecules) according to the product of interest- The ratio of substrate to acceptor molecule (sucrose to maltose) is key factor in determining nature of product formed. Additionally, alternansucrase works with multiple acceptor molecules such as raffinose, cellobiose etc. providing flexibility in manufacturing process.
- Easy downstream processing- Enzymatic synthesis is regio-selective, therefore reduces by-product formation during reaction. Thus, simple purification methods (like membrane filtration) can be used for quick and easy recovery of finished product (oligo/ polysaccharides).

The process flow chart of use and fate of alternansucrase in the production of maltose alternan oligosaccharides is shown below.

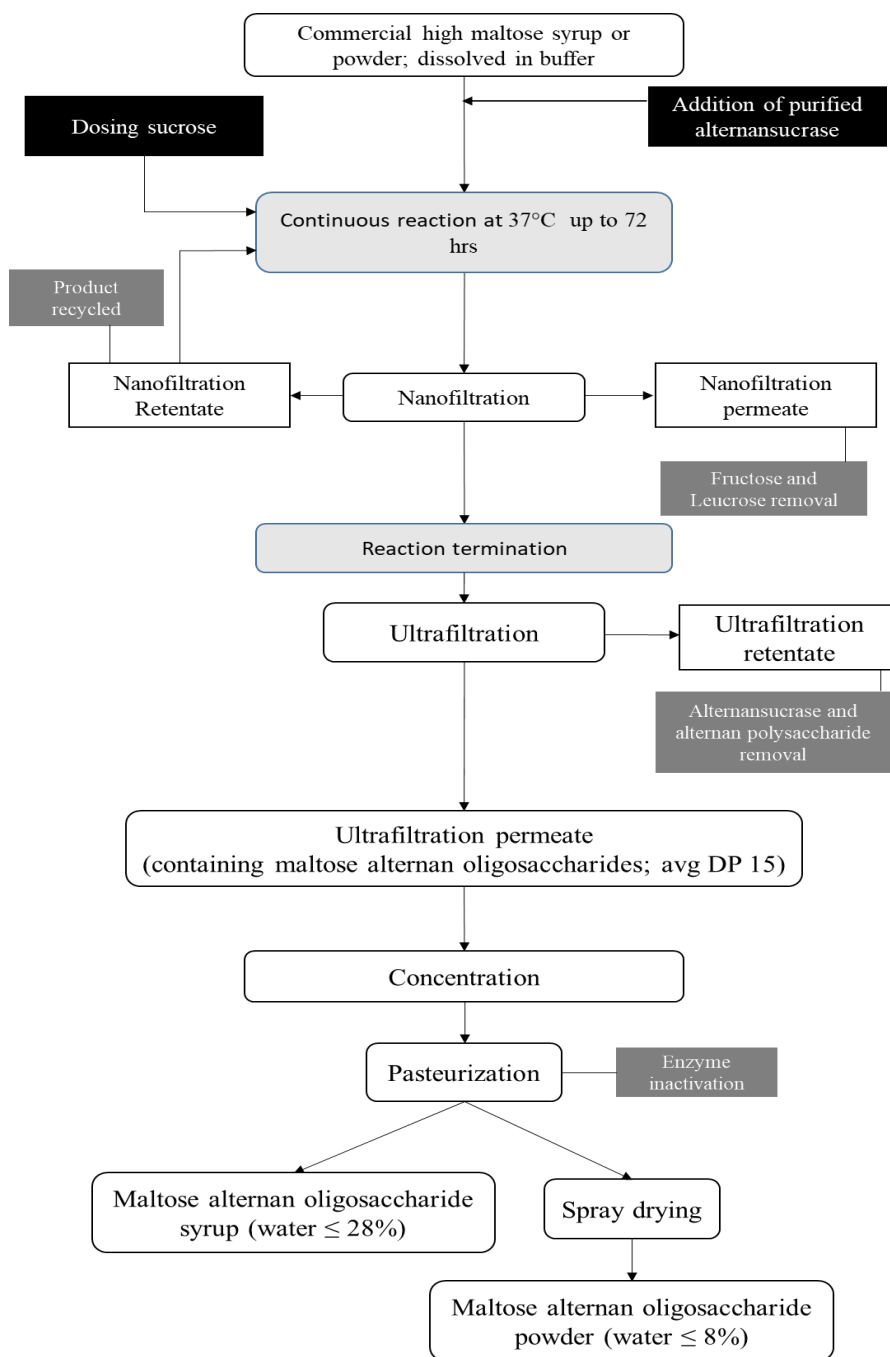


Figure. 6: Flow-chart - Use and fate of the alternansucrase in production of maltose alternan oligosaccharides

Bakery process (Bread making)

Sugar is a crucial ingredient in baking, primarily because it helps create golden brown crusts, enhances crumb texture, and retains moisture in the crumb. However, refined sugar, commonly used in baked goods, is high in calories and has a high glycemic index, offering no essential nutrients or clear health benefits (Olagunju, 2019). While processing of bakery products, alternansucrase can be added during the initial phase i.e. dough or batter mixing where it will perform its technological function and reduce the sugar content. The sucrose in the dough will be hydrolyzed by alternansucrase to release free fructose and glucose. The released glucose would be utilized for oligosaccharide and/or polysaccharide (glucans) formation. The released fructose will enhance the sweetness and make up for the flavor loss caused by the reduction in sucrose. The resultant product will be low calorie. The glucans are resistant to enzymatic digestion and possess dietary fiber potential. The glucans would also be responsible to improve bread volume and retain moisture. Further, they would act as prebiotics.

The benefits of the conversion of sucrose with the help of alternansucrase in bakery process are:

- Sucrose reduction- The sucrose content in the final food would be reduced due to the technological function of alternansucrase enzyme added in dough.
- Increase dietary fiber content- The oligosaccharides and/or polysaccharides (glucans) formed are resistant to enzymatic digestion and exhibit dietary fiber potential (Vaze et al., 2024).
- Prebiotic potential- The resulting oligosaccharides are reported to possess prebiotic properties i.e. they support growth of probiotic bacteria (Sanz et al., 2005).
- Moisture retention and mouth feel- The resulting glucans contribute to moisture retention and mouth feel.
- Improved bread volume- The resulting glucans aids in improving bread volume

The process flow chart of use and fate of alternansucrase in the bakery process is shown below.

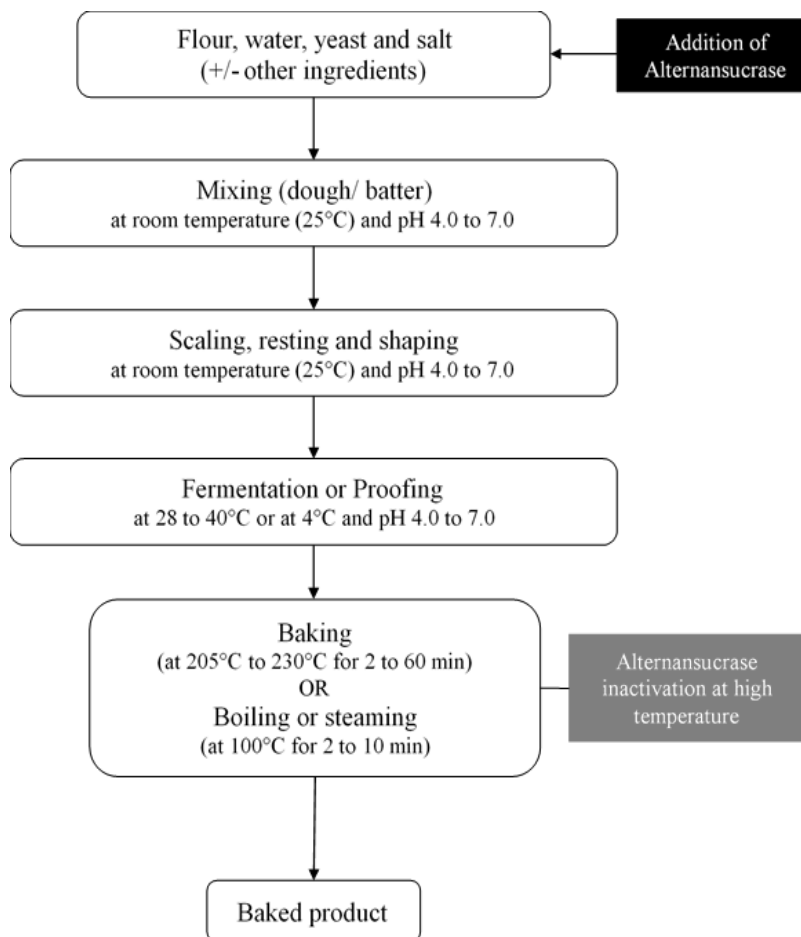


Figure. 7: Flow-chart - Use and fate of the alternansucrase in bakery product preparation

Fruit processing (Juice production)

Fruit-based products, such as juices, are valuable sources of vitamins but are also high in calories due to presence of sugars (mono and disaccharides). Recent studies indicate that high consumption of simple sugars can lead to negative health effects, including obesity, type II diabetes, and cardiovascular disease (Giri et al., 2018).

Typically, oligosaccharides are added as functional food additives to various products after being enzymatically produced from pure sugars. However, recent reports suggest a novel approach of enzymatically producing oligosaccharides using the sugars already present in food products (Vafeiadi, et al., 2016).

Alternansucrase can be added to the fruit juice to reduce its sugar content. This enzyme converts sucrose into slowly and/or partially digestible oligosaccharides/polysaccharides, collectively known as glucans. These glucans have dietary fiber properties and can act as prebiotics. The resulting fruit juice thus offers greater health and nutritional benefits.

Benefits of conversion of sucrose with the help of alternansucrase in fruit processing are:

- Low sugar content- Addition of alternansucrase in juice processing converts sugar into oligosaccharides and/or polysaccharides that are resistant to endogenous enzymatic digestion in the body. Hence, the resulting product has low sugar content.
- Increase dietary fiber content - The oligosaccharides and/or polysaccharides formed in presence of alternansucrase exhibit dietary fiber potential (Vaze et al., 2024).
- Prebiotic potential - The oligosaccharides produced by alternansucrase are known to support growth of probiotics (Sanz et al., 2005).

The process flow chart of use and fate of alternansucrase in the fruit juice production is shown below:

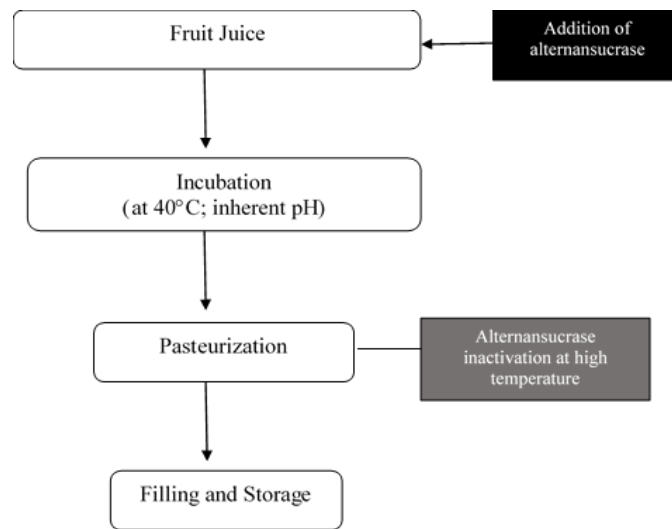


Figure. 8: Flow-chart - Use and fate of the alternansucrase in fruit juice production

2.5.3 USE LEVELS

Recommended use level of alternansucrase is given in Table 4 based on food application.

Table 4: Recommended use levels

Application	Raw material (RM)	Recommended use levels (mg /kg RM)	Recommended use levels (mg TOS/kg RM) ⁸	Maximal recommended use levels (mg TOS/kg RM)
Maltose alternan oligosaccharide production	Sucrose + Maltose	229.92-2299.32	192.72-1927.29	1927.29
Bakery process (bread making)	Flour, water, yeast, salt, sugar	130-390.22	108.97-327.08	327.08
Fruit processing (Juice production)	Fruit juice	65-260.15	54.48-218.05	218.05

The *Quantum Satis* principle is used by food manufacturers in relation to food enzyme preparations, which means that food manufacturers will typically fine-tune the enzyme dosage based on a dose range recommended by the enzyme supplier.

2.5.4 ACTIVE AND INACTIVE ENZYME RESIDUES IN THE FINAL FOOD

Alternansucrase performs its technological function during food processing but has no technical effect in final oligosaccharide food. The reasons why the alternansucrase does not exert enzymatic activity in the final food can be due to a combination of various factors, depending on the application and the process conditions used by the individual food producer. These factors include removal of the enzyme from the food during the process, denaturation of the enzyme during processing, depletion of the substrate, lack of water activity, unconducive pH, etc.

The fate of alternansucrase in production of maltose alternan oligosaccharides, baking process and fruit juice processing is explained in more details below:

Production of maltose alternan oligosaccharides:

Post the enzymatic action of alternansucrase with maltose and sucrose forming maltose alternan oligosaccharides, the physical separation of the enzyme is simple during the subsequent processing of the oligosaccharides. The alternansucrase has high molecular weight and oligosaccharides have a theoretical molecular weight far less than the alternansucrase enzyme. During the downstream processing, ultrafiltration membranes are used in the purification of the oligosaccharides, which allow the separation of the product of interest (oligosaccharides) while removing the high molecular weight enzyme from the retentate stream. Also, pasteurization performed after the concentration step inactivates any residual activity left during the process.

⁸ TOS value considered for calculation 83.82%.

Bakery process (Bread making)

Alternansucrase performs its technological function during different phases of bread making such as dough mixing, resting and/ or proofing. These steps are proceeded by baking i.e. heating or roasting dough at very high temperature (205 to 230°C); since alternansucrase is a protein it will denature during this process and will be inactivated. Therefore, active form of alternansucrase will not be present in the final food

Fruit processing (Juice production)

Alternansucrase enzyme is added to fruit juice where it performs its technological function. The fruit juice is then subjected to pasteurization where enzyme will get inactivated. Hence, active form of alternansucrase will not be present in the final food.

Part 3: 21 CFR 170.235: Intended Use and Dietary Exposure

The alternansucrase covered in this dossier is intended to be used in production of maltose alternan oligosaccharides, baking process and fruit juice production.

Due to this wide variety of applications, the most appropriate way to estimate the human consumption in the case of food enzymes is using the Budget Method (Hansen, 1966; Douglass *et al.*, 1997). This method enables to calculate a Theoretical Maximum Daily Intake (TMDI) based on certain conservative assumptions regarding 1) the level of consumption of foods and of non-milk beverages, 2) the concentration of the additive in foods and in non-milk beverages and 3) the proportion of foods and of non-milk beverages that may contain it.

Table 5: Maximal level of food enzyme alternansucrase in final food(s)

Application		Raw material (RM)	Maximal recommended use level (mg TOS/kg RM)	Example Final food (FF)	Ratio RM/FF	Maximal level in FF (mg TOS/kg food)
Solid	Baking	Flour, water, yeast, salt, sugar	327.08	Bread	0.667	218.16
	Production of maltose alternan oligosaccharides	Sucrose + Maltose	1927.29	Maltose alternan oligosaccharide	2.25	4336.40
Liquid	Fruit processing (Juice production)	Fruit juice	218.05	Juice	1.0	218.05

For the calculation of the TMDI, the maximum use levels are chosen. Furthermore, the calculation takes into account how much food or beverage is obtained per kg of raw material, and it is assumed that all the TOS will end up in the final product.

To estimate the human consumption of alternansucrase in production of maltose alternan oligosaccharides, the average dietary fiber consumption is considered in the U.S. population, and the quantity of the enzyme present in the amount of dietary fiber consumed is calculated. In this calculation, we assume a worst-case scenario where the alternansucrase preparation used in making the functional fiber is entirely carried over in the final product. This product is then consumed by the population, and all of the functional fiber is assumed to be ingested as dietary fiber.

In U.S., the recommended daily consumption of total fiber is about 21–26 g/day for adult women and 30–38 g/day for adult men, and less for children and elderly depending on age (American

Journal of Lifestyle Medicine 2017, Vol. 11 No. 1)^{9,10}. In order to calculate dietary exposure, we will consider the upper limit of total dietary fiber consumption, which is 38g/day.

Based upon the aforementioned data and assuming that all functional fiber is consumed in the form of dietary fiber, the estimated theoretical maximum daily intake (TMDI) of alternansucrase is 164.78 mg TOS/person/day¹¹. Considering the average weight of male adult to be 70 kg, the theoretical maximum daily intake (TMDI) of the alternansucrase in production of maltose alternan oligosaccharides is 2.35 mg TOS/kg body weight/day.¹²

To calculate Theoretical Maximum Daily Exposure (TMDE), the levels of consumption of foods and beverages considered are maximum physiological levels of consumption—i.e. the daily consumption of 0.1 litre of non-milk beverages per kilogram of body weight and the daily consumption of 100 kcal/kg body weight from foods (equivalent to 0.05 kg/kg body weight based on an estimated energy density of 2 kcal/g) (Hansen, 1979). The levels contained in foods and beverages are assumed to be the highest maximum levels of the additive reported in any category for foods and for beverages, respectively. The proportions of solid foods and beverages that may contain the substance are set arbitrarily. EHC 240 refers to commonly used default proportions of 12.5% for foods and 25% for non-milk beverages. (WHO EHC Chapter 6, 2009) In the case of food additives, a default proportion that is often used for European assessments is 12.5% for solid foods and 25% for beverages (WHO EHC Chapter 6, 2009).

The overall theoretical maximum daily exposure to an additive is calculated by summing the potential exposure from beverages and from foods, as shown below:

Overall theoretical maximum daily exposure = [maximum level of the additive in beverages (mg/l) × 0.1 (litre/kg body weight) × percentage of beverages that may contain the substance]
+ [maximum level of the chemical in solid foods (mg/kg) × 0.05 (kg/kg body weight) × percentage of solid foods that may contain the substance]

Consequently, the Total Theoretical Maximum Daily Exposure (TMDE) will be:

TMDI in solid food (mg TOS/kg body weight/day)	TMDI in beverage (mg TOS/kg body weight/day)	Total TMDE (mg TOS/kg body weight/day)
218.16 × 0.05 × 0.125 = 1.3635	218.05 × 0.1 × 0.25 = 5.4512	6.8147

⁹ Quagliani & Felt-Gunderson. Closing America's Fiber Intake Gap: Communication Strategies from a Food and Fiber Summit, 2017; American Journal of Lifestyle Medicine, Vol. 11, No.1

¹⁰ Institute of Medicine, Food and Nutrition Board. *Dietary Reference Intakes: Energy, Carbohydrates, Fiber, Fat, Fatty Acids, Cholesterol, Protein and Amino Acids*. Washington, DC: National Academies Press; 2005.

¹¹ Theoretical Maximum Daily Intake: 38 g of dietary fiber contains (4336.40 × 38/1000) = 164.78 mg TOS/person/day of alternansucrase preparation.

¹² 164.78 /70 kg body weight = 2.35 mg TOS/kg body weight/day of alternansucrase.

It should be stressed that this Total TMDE is based on conservative assumptions and represents a highly exaggerated value because of the following reasons:

- It is assumed that all producers of the above-mentioned foodstuffs and beverages use the specific enzyme alternansucrase.
- It is assumed that producers apply the highest use level per application;
- For the calculation of the TMDI's in food as well as in beverage, only those foodstuffs and beverages were selected containing the highest theoretical amount of TOS. Thus, foodstuffs and beverages containing lower theoretical amounts were not considered;
- It is assumed that the amount of TOS does not decrease as a result of the food production process;
- It is assumed that the final food containing the calculated theoretical amount of TOS is consumed daily over the course of a lifetime;
- Assumptions regarding food and beverage intake of the general population are overestimates of the actual average levels (Douglass *et al.*, 1997).

The NOAEL dose for alternansucrase from *E. coli* EBASSC, concluded from the 90-day toxicity study is 1000 mg TOS/kg body weight, corresponds to 3960 U/kg body weight.

The Acceptable Daily Intake (ADI) is derived from the NOAEL dose by applying a 100x safety factor, representing the highest level of consumption considered safe.¹³ The ADI for alternansucrase is 700 mgTOS/person/day¹⁴.

Based on the above calculation, the Maximum Estimated Daily Intake (EDI) is 477 mg TOS per day¹⁵, which is much lower than the ADI, i.e. 700 mg TOS per day; poses no safety concerns for consumers.

¹³ 21 CFR 170.22

¹⁴ ADI = NOAEL x 70/100, where body weight of a healthy individual is considered 70 kg and a safety factor of 100 is considered for determination of a safe dose for healthy human adults and children. (1000x 70/100= 700)

¹⁵ EDI: 6.8147 x 70kg= 477 mgTOS/day

Safety Margin

The Margin of Safety (MoS) for human consumption can be calculated by dividing the NOAEL by the TMDI value. The TMDI of the alternansucrase enzyme is 6.8147 mg TOS/kg body weight /day. Consequently, the MoS is 146.74.

$$\text{Margin of Safety (MoS)} = 1000/6.8147 = 146.74$$

As is explained above, the TMDI of the food enzyme is highly exaggerated. Moreover, the calculated NOAEL was based on the no observable adverse effect at the highest dose administered and is therefore considered as a minimum value. Therefore, the actual MoS in practice will likely be some magnitudes higher.

Based upon the data presented above, the use of alternansucrase as a processing aid does not present a toxicological concern for consumers.

Part 4: 21 CFR 170.240: Self-Limiting Levels of Use

There are no self-limiting levels of use of alternansucrase from *E. coli* EBASSC in food applications, as the enzyme should be used in accordance with good manufacturing practices.

Part 5: 21 CFR 170.245: Experience Based on Common Use in Food before 1958

The statutory basis for our conclusion of GRAS status in the notice is scientific procedures rather than on common use in food prior to 1958.

Part 6: 21 CFR 170.250: Narrative

6.1. Safety of Donor and Production Strain

The microorganism used to produce alternansucrase is a non-pathogenic, non-toxigenic strain *E. coli* EBASSC. The microorganism has been genetically modified. The recombinant production strain *E. coli* EBASSC is derived from *Escherichia coli* W3110 which is a well-characterized, non-pathogenic laboratory strain by the addition of the plasmid pAML369. *E. coli* W3110 strain was derived from the well characterized wild-type parent *E. coli* K-12, which is the most widely used laboratory strain with a history of safe use. (Gorbach, 1978, Bauer *et al.*, 2007, GRN 624). The *E. coli* B, C and W strains and their derivatives are all classified as risk group 1 organisms in biosafety guidelines (Bauer *et al.* 2007). The *E. coli* K-12 does not contain functional/intact gene sequences encoding an O antigen polysaccharide (US EPA, 1997). It has also been reported that *E. coli* K-12 does not have the well-recognized pathogenic mechanisms required by strains of *E. coli* causing the majority of enteric infections. The absence of pathogenic potential of *E. coli* K-12 is demonstrated in the feeding studies by administration of *E. coli* K-12 to GI track of animals (Freter R. *et al* 1983, Curtiss 1978). In mice, chickens, pigs and calves no pathogenic evidence of *E. coli* K-12 was observed (Gorbach 1978). Additionally, the US Environment Protection Agency (EPA) has granted tiered exemption to *E. coli* K-12, due to its extensive history of safe use. (US EPA, 1997)

E. coli K-12 has been widely used as safe laboratory organism because it doesn't produce toxins that cause illness to human by ingestion (US EPA, 1997) such as shiga-like toxins produced by some other *E. coli* strains. No records of K-12 enterotoxin-induced disease for fermentation workers were reported so far (US EPA, 1997). Also *E. coli* K-12 doesn't carry known virulence genes. Finally, *E. coli* K-12 is observed to be non-competitive and poor colonizers, making *E. coli* K-12 unlikely to inhabit the GI track of humans or other animals (Freter *et al.*, 1983).

Further, the *asr* gene donor microorganism- *L. mesenteroides* is a member of lactic acid bacteria (LAB), which has a long history of safe use in fermented food production, industrial-scale enzyme production and consumption as a probiotic and can be considered as safe for production of enzymes for food as well as feed processing and numerous other industrial applications. Importantly, EFSA has assigned *L. mesenteroides* a "Status of Qualified Presumption of Safety" (QPS) (EFSA 2007, 2020) and *L. mesenteroides* is included in the Inventory of Microbial Food Cultures with Safety Demonstration in Fermented Food Products by the International Dairy Federation (495/2018).¹⁶

6.2 Safety of the Alternansucrase Enzyme

Alternansucrase has a history of use in the food industry. Alternansucrase is a member of the glycosyltransferase family, specifically hexosyltransferases. Alternansucrase catalyzes a chemical reaction that transfers an alpha-D-glucosyl residue from sucrose alternately to the 6- and 3-positions of the non-reducing terminal residue of an alpha-D-glucan, thereby creating a glucan with

¹⁶ Bulletin of the international dairy federation 495/2018

alternating alpha-1,6- and alpha-1,3-bonds. The name "alternan" was coined in 1982 (Cote & Robyt 1982b) for the glucan based on its alternating linkage structure. The oligosaccharide acceptor products formed by this enzyme have been studied for over 20 years (Cote and Robyt, 1982b; Pelenc *et al.*, 1991) and are of interest because of their potential prebiotic activity. These oligosaccharides have also shown potential as a low-glycemic index sweetener (Carlson and Woo, 2004).

Alternansucrase from *Leuconostoc citreum* strain NRRL B-30894 has been evaluated for its safety and technological need by the European Food Safety Authority and found no safety concern under intended conditions of use. (EFSA Journal 2021b;19(1):6367)¹⁷

Sucromalt, produced by alternansucrase derived from *Leuconostoc citreum*, is approved for sale in Canada. Health Canada notified that it has no objection to the sale of sucromalt as a food ingredient that can be used as a partial or complete substitute for sweetening agents available in the Canadian market, e.g., sucrose, corn syrup, high fructose corn syrup (HFCS), brown rice syrup and malt syrup.

The toxigenic potential of the alternansucrase preparation from *E. coli* EBASSC was evaluated in animal and other cell-based toxicity model systems. The alternansucrase was found to be non-toxic, non-mutagenic, non-clastogenic and non-aneugenic under the given test conditions at the highest dose tested.

Heavy metal and mycotoxin contents in the alternansucrase produced by *E. coli* EBASSC were within the specifications and below the detection limits of the test methods. Further, the alternansucrase covered in this dossier and produced by *E. coli* EBASSC does not contain any detectable amount of full-length recombinant DNA as determined by the absence of amplicon for a full-length recombinant gene of alternansucrase. This confirmed the absence of recombinant DNA in the product (see Section 2.2.7 for details).

6.2.1 ANTIMICROBIAL ACTIVITY OF ALTERNANSUCRASE ENZYME PRODUCED BY A RECOMBINANT *E. COLI* STRAIN

The alternansucrase enzyme was evaluated for its antimicrobial activity as per JECFA (FAO/WHO, (2006b), Volume 4). The antibacterial activity was conducted against six test strains obtained from American Type Culture Collection (ATCC).

Three batches of alternansucrase were checked for their antimicrobial activity against the six selected microorganisms as recommended by FAO/WHO, (2006b), Volume 4. [*Staphylococcus aureus* ATCC 6538, *Escherichia coli* ATCC 11229, *Bacillus cereus* ATCC 2, *Bacillus circulans* ATCC 4516, *Streptococcus pyogenes* ATCC 12344, *Serratia marcescens* ATCC 14041].

¹⁷ *L. mesenteroides* is phylogenetically close to *Leuconostoc citreum*. The 16S rRNA gene sequence analysis showed that *L. mesenteroides* shares 97-99% sequence homology with *L. citreum* (Murcia and Collins, 1990) Martinez-Murcia AJ, Collins MD (1990) A phylogenetic analysis of the genus *Leuconostoc* based on the reverse transcriptase sequencing of 16S rRNA. FEMS Microbiol Lett 70:73–84

Alternansucrase enzyme preparation showed the absence of antibacterial activity against all test microorganisms.

6.2.2 **ALLERGENICITY**

The allergenic potential of the alternansucrase was assessed following the pepsin digestion and bioinformatics analyses of its amino acid sequence.

6.2.2.1 **Evaluation of allergenicity of the alternansucrase following pepsin digestion method**

Allergenicity of the alternansucrase from *E. coli* EBASSC was evaluated following pepsin digestion method. The pepsin digestion study is based on the recommendations made by the Joint FAO/WHO Expert Consultation, 2001. The protocol used for this assay is based on the detailed study undertaken by Thomas et al. (2004), which in turn takes into consideration the guidelines provided by the Codex Alimentarius Commission, 2003.

Proteins that are susceptible to digestion by pepsin are less likely to elicit an allergenic response. Hence, the digestibility of the alternansucrase by pepsin, was evaluated *in vitro* by exposing it to the action of pepsin in the presence of simulated gastric fluid (SGF).

Results from this study showed that alternansucrase can be considered to be a labile protein, as it is digested by pepsin, and therefore is not likely to be allergenic.

6.2.2.2 **Sequence homology of the alternansucrase protein with the known allergens**

Allergenic potential of the alternansucrase was assessed following the bioinformatics analyses of its amino acid sequence. Over the last decade, bioinformatics methods have been widely used for collecting, storing, and analyzing molecular and/or clinical information of importance for determining the allergenic potential of proteins. In order to address the allergenicity potential of alternansucrase, information obtained from bioinformatics, coupled with experimental data, wherever necessary, is the approach postulated by the Joint Food and Agriculture Organization and World Health Organization (FAO/WHO) Expert Consultation on Allergenicity of Foods Derived from Biotechnology. This approach considers the evidence derived from several types of information and data since no single criterion is considered to be sufficiently predictive (Report of a Joint FAO/WHO Expert Consultation on Allergenicity of Foods Derived from Biotechnology, 2001).

As proposed in the FAO/WHO Report, cross-reactivity between a query protein and a known allergen has to be considered when there is: (a) more than 35% identity in the amino acid sequence of the enzyme, using a window of 80 amino acids and a suitable gap penalty, or (b) A stretch identity of 6-8 contiguous amino acids.

The alternansucrase sequence was analyzed using the Structural Database of Allergenic Proteins (SDAP) Program database.¹⁸ Bioinformatics searches showed that alternansucrase from *E. coli*

¹⁸ Food Allergy Research and Resource Program <https://farrp.unl.edu/>

EBAASC, showed a maximum of 30.00%, identity with Allergen Asp f 17 and Allergen Poa p a. On conducting a full length FASTA alignment, the PID was less than 50%. Additionally, no hits were obtained for a match of 8 contiguous amino acids. This analysis, based on the comparison of the alternansucrase with the amino acid sequences of known allergen, showed no allergenicity concern.

6.2.2.3 Comprehensive literature search and discussion on the possibility of the allergic reaction

Enzymes are proteins in nature. Like many other proteins, they may have the potential to cause allergic responses. Allergens can activate the immune system through several routes of exposure, including food, skin contact, and inhalation. Enzymes have been reported to cause allergies through recurrent exposure via inhalation in sufficiently high doses and when they come into contact with mucosal surfaces (eyes, nose). Only a few instances of enzyme-induced allergies have been documented for enzyme allergy development from contact with eyes (Griffith et al. 1969; Greenough et al. 1991). Cases of respiratory allergy / sensitization following occupational inhalation of enzymes have been reported. (Vanhanen M. 2001; Bernstein et al., 1994).

Alternansucrase is a member of the glycosyltransferase family, in particular they are hexosyltransferases. There are very few cases been reported so far stating allergic response triggered by glycosyltransferase family members, particularly by alternansucrase. One case report was published on oral allergy syndrome due to glycosyltransferases. In this case report, they claimed that glycosyltransferases expressed in custard apple acts as an allergen when consumed orally and it triggers IgE mediated allergic response with oral allergy syndrome. (Carrón-Herrero A et al, 2022).

Alternansucrase obtained from other *Leuconostoc spp. i.e., Leuconostoc citreum* was assessed for allergenicity potential and showed no safety concern. EFSA has published a positive scientific opinion for alternansucrase enzyme from *Leuconostoc citreum* NRRL B-30894. The EFSA panel has estimated that risk of allergic sensitization and elicitation reactions upon dietary exposure to this food enzyme is considered low (EFSA Journal 2021b;19(1):6367)¹⁹.

The donor organism *Leuconostoc mesenteroides* is a member of lactic acid bacteria (LAB) group and *Leuconostoc* genus has received the status of Qualified Presumption of Safety (QPS) by the European Food Safety Authority (EFSA 2007, 2020). *Leuconostoc* members are widely known for their roles in several industrial and food fermentation processes (Buckenhushkes, 1993; Steinkraus, 2002) such as fermented sausages, fermented vegetables and cereal products, and dairy products (e.g., butter, cream, fresh and raw milk, cheeses). This abundant presence of *Leuconostoc* strains in various foods and the absence of a linkage between the fermented foods and allergies makes it highly

¹⁹ *L. mesenteroides* is phylogenetically close to *Leuconostoc citreum*. The 16S rRNA gene sequence analysis showed that *L. mesenteroides* shares 97-99% sequence homology with *L. citreum* (Murcia and Collins, 1990) Martinez-Murcia AJ, Collins MD (1990) A phylogenetic analysis of the genus *Leuconostoc* based on the reverse transcriptase sequencing of 16S rRNA. FEMS Microbiol Lett 70:73–84

unlikely that enzymes contained in these organisms are responsible for any allergic sensitization in consumable foods.

The microorganism used for the production of alternansucrase is a derivative of non-pathogenic, nontoxigenic strain of *E. coli* K-12. Bioinformatic observations have demonstrated that *E. coli* K-12 does not contain functional/intact gene sequences encoding an O antigen polysaccharide (US EPA, 1997). No evidence has been reported so far related to allergic response triggered due to non-pathogenic strains of *E. coli* although in few cases *E. coli* strains have been used for attenuation of allergic response. Oral administration of the probiotic strain *E. coli* (Nissle 1917) (ECN) helps in prevention of papain induced respiratory barrier injury and severe allergic inflammation in mice (Secher et al. 2018).

In general, there are no reports of sensitization to the enzyme products in the final commercial food after ingestion. This may be due to the difference in exposure pattern of the inhalation route compared to the digestive route, or it may be due to the fact that the enzymes, again generally, most often are present in the final foods in low amounts and in inactive forms (Bindslev et al 2006). Enzyme proteins used in food are not homologous to known food allergens.

The alternansucrase is not, itself, a known allergen. Alternansucrase is used in a very small amounts during food processing and removed from the oligosaccharide food during processing, resulting in miniscule quantities of the enzyme in the oligosaccharide or foods contain them. Goodman et al (2008) discuss that a high concentration generally equals a higher risk of sensitization, whereas a low level in the final food equals a lower risk. Additionally, it must be noted that the food enzyme protein may undergo denaturation under various conditions of food processing, resulting in loss of its tertiary structure. In general, these alterations in conformation are associated with a decrease in the antigenic reactivity in humans. Usually, denatured proteins are much less immunogenic than the corresponding native proteins (Takai et al., 1997; Valenta, 2012; Kikuchi et al., 2006). Additionally, residual enzyme(s) still present in the final food will be subjected to digestion in the gastro-intestinal system, which further reduces the risk of enzyme allergenicity.

Further, materials used in the fermentation process (inoculum, seed, and main fermentation) of alternansucrase from *E. coli* EBASSC are food-grade substances that are not known allergens and are not one of the listed allergens of concern (FASTER Act, 2021)²⁰ Moreover, no allergenic ingredients are added post fermentation in enzyme preparation, causing no allergenicity or labelling concerns.

In summary, the alternansucrase enzyme is readily digested by the pepsin, is not homologous to know allergens, is present in the final food in very low amounts, is present in a microbial food culture that is not known to elicit allergenic reactions in consumers of the food and is produced in an organism similarly not known to elicit allergenic responses. Bioinformatics analyses do not show any similarity of alternansucrase amino acid sequence with a known allergen. Based upon the

²⁰ Food Allergy Safety, Treatment, Education, and Research Act of 2021 ([FASTER Act of 2021](#))

preponderance of evidence, it is highly unlikely that the alternansucrase enzyme presents allergenicity concerns as a result of the intended uses.

6.2.3 **REGULATORY HISTORY AND LEADING ENZYME PUBLICATIONS ON SAFETY OF THE ENZYME**

Alternansucrase has a safe history of use in food processing. Foods processed with alternansucrase are known to be consumed safely by the general human population.

European Food Safety Authority has evaluated alternansucrase from non-genetically modified *Leuconostoc citreum* strain NRRL B-30894 as a processing aid and found no safety concern under intended conditions of use (EFSA Journal 2021b;19(1):6367)²¹.

6.3 **Safety of Manufacturing Process**

The alternansucrase described in this dossier is manufactured under current Good Manufacturing Practice (cGMP) in accordance with 21CFR§110 and 21CFR§117 and the principles of Hazard Analysis and Critical Control Points (HACCP). The manufacturing facility is ISO 9001, ISO 22000 (Food Safety Management System) and GMP certified. The analysis of three (3) non-consecutive batches demonstrates that the process is capable of producing product that meets the established specifications.

6.4 **Safety of Alternansucrase —Oral Toxicity and Genotoxicity Studies**

Alternansucrase from *E. coli* EBASSC, the notified enzyme, has been investigated in a series of systemic and genetic toxicity studies complied with OECD Guidelines and conducted according to the principles of Good Laboratory Practice (GLP) as published by the OECD (ENV/MC/CHEM (98)17). Refer to the Rathi et al. 2023 article for a detailed description of toxicity study on alternansucrase.

Acute oral toxicity test (OECD Test No. 423, 2001):

The aim of this study was to estimate LD₅₀ for alternansucrase and classification of test item according to the GHS system for chemicals causing acute toxicity.

Method:

In this study, single oral administration of test item was made to groups of three female Wistar rats in stepwise manner to assess its acute toxicity.

²¹ *L. mesenteroides* is phylogenetically close to *Leuconostoc citreum*. The 16S rRNA gene sequence analysis showed that *L. mesenteroides* shares 97-99% sequence homology with *L. citreum* (Murcia and Collins, 1990) Martinez-Murcia AJ, Collins MD (1990) A phylogenetic analysis of the genus *Leuconostoc* based on the reverse transcriptase sequencing of 16S rRNA. FEMS Microbiol Lett 70:73–84

In step-1 of the study, the test item was administered to three female rats, at the starting dose of 300 mg/kg body weight. The test item did not induce any clinical signs and mortality in treated rats. The body weight gain by treated rats was not affected during the 14 days observation period post treatment. No gross pathological changes were observed in any of the rats, as evident at terminal necropsy.

In step-2, as the first dosed animals survived, the test item was further tested in another group of three rats at the dose of 300 mg/kg body weight. The test item did not induce any abnormal clinical signs and mortality in treated rats. The body weight gain by treated rats was not affected during the 14 days observation period post treatment. No gross pathological changes were observed in any of the rats, as evident at terminal necropsy.

During step-3 and step-4, the test item further tested at the dose of 2000 mg/kg body weight and observed for incidence of death and signs of toxicity for a period of 14 days.

Observations/Results:

No signs of toxicity or mortality were observed in all six female rats throughout the study period of 14 days when treated at the dose level of 2000 mg /kg of the test item. The test item did not induce any abnormal clinical signs in treated rats. The body weight gain by treated rats was not affected during the 14-day observation period post treatment. Gross pathological examination did not reveal any abnormalities in animals from the 2000 mg /kg dose group.

Conclusion:

Based on these results, the estimated LD₅₀ for alternansucrase is greater than 2000 mg /kg body weight. It was concluded that the acute toxicity study of alternansucrase; when administered via oral route in Wistar rats fall into the category 5 criteria of Globally Harmonized System (GHS). Alternansucrase is classified in Globally Harmonized System (GHS) category 5 or unclassified for the obligatory labelling requirement for oral toxicity, the corresponding LD₅₀ value lying between $< 2000 < ATE \leq 5000$ or > 5000 mg/kg body weight respectively. The LD₅₀ cut-off value of Alternansucrase is 5000 mg/kg body weight.

Repeated-dose 90-day oral toxicity test (OECD Test No. 408, 2018):

The aim of this study was to investigate the clinical, hematological and histological toxicity potential of alternansucrase (Batch No. 122101), following a 90-days oral administration in male and female rats receiving a daily treatment.

Method:

Alternansucrase was administrated to animals at the dose level of 250 mgTOS/kg, 500mgTOS/kg and 1000 mgTOS/kg body weight up to day 90. Two additional dose levels were added to the study as 0 mgTOS/kg (Rev.) and 1000 mgTOS/kg (Rev.), in order to study the reversibility or delayed occurrence of symptoms, if any. The control animals were administrated with vehicle (distilled water) only. The doses were selected based on results of the acute toxicity study and 14 days dose range finder study. The rats were examined daily for signs of toxicity, morbidity and mortality.

Result/Observation:

There was no incidence of mortality or abnormal clinical signs in rats treated at and up to the dose of 1000 mgTOS/kg during the treatment and recovery period. There was no incidence of ophthalmic abnormalities in rats of high dose groups treated at and up to the dose of 1000 mgTOS/kg of body weight. Also, there were no ophthalmic abnormalities in rats of control group.

The observations on sensory reactivity, grip strength and motor activity conducted during last week of treatment, did not reveal any neurotoxic potential of the test item alternansucrase at and up to the dose of 1000 mgTOS/kg body weight.

No effect on body weights, body weight gain and food consumption were observed in male and female rats treated with test item alternansucrase at and up to the dose of 1000 mgTOS/kg and were found to be comparable to those of control rats throughout the treatment and recovery period. No test item related adverse effect observed on estrus cycle of female rats based on the vaginal smear examination performed at the end of the treatment period and also at the end of the recovery period.

There was no adverse effect on hematology and coagulation parameters in male and female rats treated at the dose of 250 mg TOS/kg body weight. The changes in neutrophils, lymphocytes, and monocyte count in rats treated at 500 mg TOS/kg & 1000 mg TOS/kg were considered non-adverse due to reversal of changes in the recovery group. The group mean values of hematology parameters (except neutrophils, lymphocytes and monocyte count in male rats) of rats, treated with alternansucrase at and up to the level of 1000 mgTOS/kg of body weight were found to be comparable to rats of control group at the end of the treatment period and also at the end of the recovery period.

The group mean values of clinical chemistry and urinalysis parameters of male and female rats, treated with alternansucrase at and up to the level of 1000 mgTOS/kg of body weight were found to be comparable to rats of control group at the end of the treatment period and also at the end of the recovery period. In male rats, test item induced mild increase was observed in thyroxine (T4) levels treated at 1000 mgTOS/kg and triiodothyronine (T3) levels treated at 500 mgTOS/kg and 1000 mgTOS/kg. Increase in thyroid stimulating hormone (TSH) was considered as toxicologically insignificant due to the lack of correlating findings in other thyroid hormone levels, no changes in absolute and relative weights of thyroid gland and no gross and histopathological alteration in thyroid gland. Also, reversal was observed in the levels of thyroxine (T4), triiodothyronine (T3) as well as in the absolute and relative thyroid gland weights at the end of recovery period.

The values of absolute and relative weights of kidneys, liver, adrenals, testes, epididymides, uterus including cervix, thymus, spleen, brain, ovaries, heart, pituitary gland, prostate and seminal vesicles with coagulating glands of rats treated with alternansucrase at and up to the dose level of 1000 mgTOS/kg body weight were found to be comparable with those of the control rats at the end of treatment period and also at the end of recovery period. Increase in absolute and relative thyroid weight (fixed) was observed in male rats treated at 1000 mgTOS/kg body weight at the end of treatment period but considered as non-adverse due to its reversal at termination of recovery period.

No test item related gross and histopathological changes were observed in any of the rats treated with alternansucrase.

Conclusion:

Based on the findings The No Observed Adverse Effect Level (NOAEL) of alternansucrase in Wistar rat via oral route, over a period of 90 days was found to be 1000 mgTOS/kg body weight in male and female animals. This corresponds to 3960 U/kg/day for male & female rats.

Bacterial reverse mutation test—Ames assay (OECD Test No. 471, 1997):

The aim of this *in-vitro* study was to detect the possible presence of mutagenic compounds or contaminants in the test substance, alternansucrase from *E. coli* EBASSC (Batch no. 122101). This test enables the detection of potential mutagens which act by causing base-pair substitutions or frameshift mutations. Tester strains of *Salmonella typhimurium* are used in this study).

Method:

Initially, the dose range finding study was conducted to evaluate the cytotoxicity of the test item.

Based upon the preliminary tests conducted to assess the solubility/ precipitation and cytotoxicity of alternansucrase, the tester strains were exposed to the test item in triplicate cultures at the concentrations given below.

Tester Strains	Concentration (µg TOS/plate)					
	Highest Dose	Lower doses				
With and Without Metabolic Activation (S9)						
TA97a, TA98, TA100, TA102 and TA1535	5000	1500	500	150	50	15

Liver S9, induced in Wistar rats with sodium phenobarbitone and β -naphthoflavone, was used as metabolic activation system for the experiments. Analytical grade water was used as a vehicle. The exposed bacteria were plated onto minimal glucose agar medium supplemented with L-histidine. The plates were incubated at 37°C for about 66 to 67 hours, after which the histidine revertant colonies were counted and their frequency was compared with that in the vehicle control group. Concurrent positive control groups were also included in the experiments, as specified by the test guideline.

Result/Observation:

Plate counts for the spontaneous histidine revertant colonies in the vehicle control groups were found to be within the frequency ranges expected from the laboratory historical control data at INTOX PVT. LTD. They also compared well with the range reported in the literature.

Concurrent positive controls demonstrated sensitivity of the assay with and without metabolic activation.

Results of this test indicated that the frequencies of histidine revertant colonies at all concentrations of alternansucrase in tester strains TA1535, TA97a, TA98, TA102 and TA100 in the presence and absence of a metabolic activation system, were comparable to those observed in the vehicle control group and this observation was confirmed by repetition of the experiment.

Conclusion:

Based on these results, it is concluded that alternansucrase tested at and up to 5000.00 µg TOS/plate concentration did not induce any mutations in the presence and absence of microsomal enzymes (S-9 fraction) and is therefore non-mutagenic in this Bacterial Reverse Mutation Test.

***In vitro* mammalian cell micronucleus test (OECD Test No. 487, 1997):**

The aim of this *in vitro* mammalian cell micronucleus test using Human Lymphoblastoid Cell Line - TK6 was to evaluate the potential of alternansucrase to induce formation of small membrane-bound DNA fragments (micronuclei) in the cytoplasm of interphase cells. These micronuclei may originate from acentric fragments (chromosome fragments lacking a centromere) or whole chromosomes that are unable to migrate with the rest of the chromosomes during the anaphase of the cell division. The test has the potential to detect the activity of both clastogenic and aneugenic chemicals.

Method:

Test concentrations for this study were selected based on the results obtained from a solubility and precipitation test of the test item in the final treatment mixture, followed by a preliminary cytotoxicity test. Based on these results, TK6 cells were exposed to the test item at concentrations of 1250, 625 and 312.5 µg TOS/ml with and without metabolic activation for 3 hours and 24 hours exposure. Liver S9, induced in Wistar rats with sodium phenobarbitone and β-naphthoflavone, was used as metabolic activator system for the experiments. Triplicate cultures were used at each concentration.

Micronuclei scoring was performed using flow cytometric method (Litron-MicroFlow).

In experiment no. 1, cells were exposed to the test item for about 3 hours in presence of metabolic activation system, cell harvesting was done after about 1.5-2 cell cycle length after the beginning of treatment.

Testing in absence of metabolic activation system was conducted as experiment no. 2 and 3, in which the exponentially proliferating cells were exposed to the test item for 3 hours and 24 hours respectively. Cell harvesting was done after about 1.5-2 cell cycle length after the beginning of treatment.

Positive and vehicle controls, both with and without metabolic activation were tested concurrently with the test item. Analytical grade water was used as a vehicle control. Known micronucleus forming (aneugenic and clastogenic) agents were employed as positive controls and included mitomycin C and vinblastine, at the concentrations of 160 ng/ml and 3.5 ng/ml respectively, for

experiment without metabolic activation system, while cyclophosphamide was employed at the concentration of 6.25 µg/ml for experiment with metabolic activation system.

At the end of treatment/recovery period, cells were lysed, stained and about 10,000 healthy nuclei per group were analyzed using flow cytometer. A comparison of the % incidence of micronuclei for each of the three experiments conducted in this study, either with or without metabolic activation system.

Positive Controls: Sensitivity of the test system and activity of S9 mix were demonstrated in the positive control group. The positive controls, viz. directly acting clastogen mitomycin C, directly acting aneugen vinblastine and indirectly acting clastogen cyclophosphamide, induced significant increase in frequencies of micronuclei over the concurrent controls, which validated the test method.

Cytotoxicity: Test item induced 8.13, 6.15 and 6.93% cytotoxicity to the TK6 cells at the highest test concentration in experiment no. 1, 2 and 3 respectively in main study.

Result/Observation:

The test item alternansucrase did not induce significant increase in the incidence of micronuclei over the tested range, in any of the three experiments conducted, both with and without metabolic activation system.

Conclusion:

Based on these results it is concluded that the test item, alternansucrase was found to be non-clastogenic and non-aneugenic at and up to the test concentration of 1250 µg TOS/mL.

Based on the toxicity data, we conclude that the alternansucrase enzyme preparation does not exhibit adverse toxicological effects under the experimental conditions described.

6.5 Decision Tree

The safety of alternansucrase from *E. coli* EBASSC has also been established using the decision tree for evaluating the safety of microbial enzyme preparations used in food processing (Pariza et al., 2001). The decision tree is a widely used decision making tool by the enzyme industry and has been utilized for the purpose of GRAS determinations that have been reviewed by FDA.

1. Is the production strain genetically modified? **YES**

If yes, go to 2. If no, go to 6.

2. Is the production strain modified using rDNA techniques? **YES**

If yes, go to 3. If no, go to 5.

3a. Does the expressed enzyme product which is encoded by the introduced DNA have a history of safe use in food? **YES,**

The alternansucrase under study is not new for food processing. Its protein sequence is not similar to known sequences of food allergens. In addition, the enzyme will be inactivated in the food manufacture process.

If yes, go to 3c. If no, go to 3b.

3c. Is the test article free of transferable antibiotic resistance gene DNA? **YES.**

The test article is free of recombinant DNA coding for the transferable antibiotic resistance gene DNA under the conditions tested. The limit of detection of the method of analysis was 10 ng of DNA per gram of the enzyme sample.

If yes, go to 3e. If no, go to 3d.

3e. Is all other introduced DNA well-characterized and free of attributes that would render it unsafe for constructing microorganisms to be used to produce food-grade products? **YES**

The genetic modifications are well characterized, and the incorporated DNA does not encode and express any known harmful or toxic substances.

If yes, go to 4. If no, go to 12.

4. Is the introduced DNA randomly integrated into the chromosome? **NO,**

The introduced DNA (alternansucrase gene) is present in the plasmid, which is extrachromosomal.

If yes, go to 5. If no, go to 6.

6. Is the production strain derived from a safe lineage, as previously demonstrated by repeated assessment via this evaluation procedure? **YES**

The production strain is a derivative of *E. coli* W3110, which is derived from the well characterized wild-type parent *E. coli* K-12.

If yes, the test article is ACCEPTED.

6.6 Safety Assessment and GRAS Determination

This section presents an assessment that demonstrates that the intended use of alternansucrase from *E. coli* EBASSC is safe and is GRAS based on scientific procedures.

This safety assessment and GRAS determination involve two steps. In the first step, the safety of the intended use of alternansucrase from *E. coli* EBASSC is demonstrated. Safety is established by demonstrating a reasonable certainty that the exposure of consumers to alternansucrase under its intended conditions of use is not harmful. In the second step, the intended use of alternansucrase from *E. coli* EBASSC is determined to be GRAS by demonstrating the safety of production organism. The enzyme under its intended conditions of use is generally recognized among qualified scientific experts and is based on publicly available and accepted information.

The regulatory framework for establishing whether the intended use of a substance is GRAS, in accordance with Section 201(s) of the Federal Food Drug and Cosmetic Act, is set forth under 21 CFR §170.30. This regulation states that general recognition of safety may be based on the views of experts qualified by scientific training and experience to evaluate the safety of substances directly or indirectly added to food. A GRAS determination may be made either: 1) through scientific procedures under §170.30(b); or 2) through experience based on common use in food, in the case of a substance used in food prior to January 1, 1958, under §170.30(c). This GRAS determination employs scientific procedures established under §170.30(b).

A scientific procedure GRAS determination requires the same quantity and quality of scientific evidence as is needed to obtain approval of the substance as a food additive. In addition to requiring scientific evidence of safety, a GRAS determination also requires that this scientific evidence of safety be generally known and accepted among qualified scientific experts. This “common knowledge” element of a GRAS determination consists of two components:

1. Data and information relied upon to establish the scientific element of safety must be generally available; and
2. There must be a basis to conclude that there is a consensus among qualified experts about the safety of the substance for its intended use.

The criteria outlined above for a scientific-procedures GRAS determination are applied below in an analysis of whether the intended use of alternansucrase from *E. coli* EBASSC is safe and is GRAS.

6.6.1 EVIDENCE OF SAFETY

As noted in the sections above, the production organism *E. coli* EBASSC, as well as the enzyme alternansucrase derived from it, are well characterized and safe for intended usage. Publicly available authorizations by various regulatory authorities, safety evaluations published in the peer-reviewed scientific literature, reviews by qualified experts, corroborated by Advanced Enzymes’ unpublished safety studies, support such a conclusion.

The microorganism used for the production of alternansucrase is a non-pathogenic, non-toxigenic strain *E. coli* EBASSC. The production strain does not produce secondary metabolites of

toxicological concern to the human. The production strain *E. coli* EBASSC is generated from the host strain *E. coli* W3110 which is a derivative of *E. coli* K-12, a non-pathogenic laboratory strain. *E. coli* K-12 is considered as safe to use for various biotechnological applications as it does not produce toxins harmful for ingestion. *E. coli* K-12 has been widely used in production of food ingredients, specialty chemicals and human drugs, because of its history of safe use it has been exempted from Environment Protection Agency (EPA), USA review of Toxic Substances Control Act, 1967 (TSCA) (EPA, 1997). Most common therapeutic proteins expressed in *E. coli* are Human insulin (Nilsson et al, 1996), interleukin 2 (IL-2) (Roifman, 1985), etc. The *E. coli* B, C, and W strains and their derivatives are all classified as risk group 1 organisms in biosafety guidelines (Bauer et al., 2007).

The alternansucrase produced by the *E. coli* EBASSC was evaluated for antimicrobial activity and the absence of antimicrobial activity was recorded against all the test micro-organisms. In addition, heavy metals in the enzyme preparations were well within the specification and below the detection limits of the analytical methods. No indications of toxicity were found in acute and repeated-dose studies of oral toxicity or genotoxicity assays of alternansucrase obtained from *E. coli* EBASSC. Pepsin digestion analysis and bioinformatics studies on alternansucrase from *E. coli* EBASSC showed that it was a labile protein indicating a reduced risk of sensitization and did not share homology with known allergens. Therefore, alternansucrase does not pose a concern in terms of food allergy induction.

Further, the alternansucrase as described in this dossier and produced by *E. coli* EBASSC does not contain the production microorganism and any detectable amount of full-length recombinant DNA as determined by the absence of amplicon for full-length recombinant gene of alternansucrase. The absence of this DNA in the final product eliminates the risk of unintended gene transfer.

The alternansucrase described in this dossier is manufactured following current Good Manufacturing Practice for Food (cGMP) and the principles of Hazard Analysis of Critical Control Points (HACCP).

Using the decision tree of Pariza and Johnson (2001), alternansucrase from *E. coli* EBASSC enzyme preparation was determined to be acceptable for the proposed uses. Based on a highly conservative (and highly unlikely) worst-case scenario, the maximum daily intake is 477 mg TOS/day which is lower than the Acceptable Daily Intake (ADI) derived from the NOAEL from the 90-day sub-chronic oral toxicity study i.e., 700 mg TOS per day. Therefore, the use of alternansucrase in the production of maltose alternan oligosaccharides, bakery process and fruit juice production is not expected to result in adverse effects in humans. A safety margin of 146.74 exists between the established NOAEL and the estimated worst-case maximum daily human exposure level.

Based upon the totality of the evidence presented in this notice, Advanced Enzyme has determined by scientific procedures that the alternansucrase from *L. mesenteroides* produced by *E. coli* EBASSC and manufactured according to the principles of cGMP is Generally Recognized As Safe for the intended processing aid uses in food.

6.7 GRAS Panel Review

The GRAS Panel, independently and collectively, critically evaluated the information and data presented in a dossier prepared by Advanced Enzymes Technologies Ltd. and summarized above and unanimously conclude that Advanced Enzyme Technologies Ltd.'s alternansucrase from *Leuconostoc mesenteroides* expressed by *Escherichia coli* EBASSC, manufactured in a manner that is consistent with current Good Manufacturing Practice for the production of food and meeting appropriate food-grade specifications, is safe and suitable and Generally Recognized as Safe (GRAS) based on scientific procedures for use as a processing aid in food production processes.

The Panel further concluded that other experts qualified to assess the safety of food ingredients would concur with these conclusions (GRAS Panel Report, Annex C).

Part 7:21 CFR 170.255: List of supporting data and information

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7.2 Annexes

Annex	Description	Reference	Pages
<u>Annex A</u>	Heavy Metals Analysis – TUV Nord Group	0122101274 0122101276 0122101278	1-6
<u>Annex B</u>	Allergen Statement	-	1
<u>Annex C</u>	Report of the GRAS Panel on the Generally Recognized as Safe (GRAS) Status of the intended uses of the <i>Leuconostoc mesenteroides</i> enzyme Alternansucrase expressed by <i>Escherichia coli</i> EBASSC	-	4

TEST REPORT**Report No : TUV(I)/10455/21-22/0122101274 PII****Date : 13 Dec 2021**

Name & Address of Customer : Advanced Enzymes Technologies Ltd.
Plot No A 61/62, Malegaon MIDC, Sinnar,,Nashik
Pin Code: 422113

Reg No. : 10455/21-22

CA No. : 0122101274

Date of sample receipt : 08 Dec 2021

Date(s) of analysis : 09 Dec 2021 - 13 Dec 2021

Sample Drawn by : Customer

SINo	Test Name	Results	Unit	LOQ	Test Method
Sample Name : Alternansucrase Batch No.: 122101 Discipline : Chemical					
CA No : 0122101274 Date of Manufacture: December 2021 Product category : Miscellaneous					
	Heavy Metals				
1	Arsenic	0.71	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
2	Cadmium	N.D.	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
3	Lead	N.D.	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
4	Mercury	N.D.	mg/kg	0.025	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition

LOQ-Limit of Quantification, N.D – Not Detected.

Authorized by



Atulkumar Rajage
Manager – Instrumentation Department

Note - This is Electronically Generated Report Copy

TEST REPORT

Report No : TUV(I)/10455/21-22/0122101274 PII

Date : 13 Dec 2021

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Plot No A 61/62, Malegaon MIDC, Sinnar,,Nashik
Pin Code: 422113

Reg No. : 10455/21-22

CA No. : 0122101276

Date of sample receipt : 08 Dec 2021

Date(s) of analysis : 09 Dec 2021 - 13 Dec 2021

Sample Drawn by : Customer

SINo	Test Name	Results	Unit	LOQ	Test Method
Sample Name : Alternansucrase Batch No.: 122103 Discipline : Chemical					
CA No : 0122101276 Date of Manufacture: December 2021 Product category : Miscellaneous					
	Heavy Metals				
1	Arsenic	0.48	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
2	Cadmium	N.D.	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
3	Lead	N.D.	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
4	Mercury	N.D.	mg/kg	0.025	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition

LOQ-Limit of Quantification, N.D – Not Detected.

Authorized by



Atulkumar Rajage
Manager – Instrumentation Department

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TEST REPORT

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10. For Pesticide Residue analysis, reported value at LOQ level may vary within analytical variation of 50% considering Uncertainty of Measurement.

-- End of Report --



TEST REPORT

Report No : TUV(I)/10455/21-22/0122101278 PII
Date : 13 Dec 2021

Name & Address of Customer : Advanced Enzymes Technologies Ltd.
Plot No A 61/62, Malegaon MIDC, Sinnar,,Nashik
Pin Code: 422113

Reg No. : 10455/21-22

CA No. : 0122101278

Date of sample receipt : 08 Dec 2021

Date(s) of analysis : 09 Dec 2021 - 13 Dec 2021

Sample Drawn by : Customer

SINo	Test Name	Results	Unit	LOQ	Test Method
Sample Name : Alternansucrase Batch No.: 122105 Discipline : Chemical					
CA No : 0122101278 Date of Manufacture: December 2021 Product category : Miscellaneous					
	Heavy Metals				
1	Arsenic	0.69	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
2	Cadmium	N.D.	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
3	Lead	N.D.	mg/kg	0.1	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition
4	Mercury	N.D.	mg/kg	0.025	TUV/03/SOP/004 Based on AOAC 2015.01, 21 st Edition

LOQ-Limit of Quantification, N.D – Not Detected.

Authorized by



Atulkumar Rajage
Manager – Instrumentation Department

Note - This is Electronically Generated Report Copy

TEST REPORT

Report No : TUV(I)/10455/21-22/0122101278 PII

Date : 13 Dec 2021

Terms and Conditions

1. Test Results are based on & related only to the particular sample(s) tested.
2. This Report cannot be re-produced, except when in full, without the written permission from TUV India Pvt. Ltd.
3. This Certificate reflects our findings at the time and place of testing.
4. Sample(s) will be retained by us for a period of one month for non-perishable items only. Perishable items will be destroyed after completion of tests.
5. This Report, in full or in part, shall not be used to make any misleading claims or for any legal purposes. For complete report refer PI
6. All terms and conditions of our quotation on the basis of which this testing service has been provided are deemed to be fully accepted by the customer and are deemed to be in full force and effect.
7. This Report is exclusively for the use of the customer whose name and address is indicated above. No third party can derive rights against the company on the basis of this report. No third party has any right to raise any claims on the company.
8. For Biological and mycotoxin Analysis: Our analytical findings reflect the quality of the sample at the time of testing. No responsibility can be accepted for the possible consequences of further development of micro-organisms or mycotoxin which may depend upon storage, handling & weathers conditions which may influence the results at a later date/time respectively.
9. The laboratory tests are conducted by TUV India as per mutual agreement without any implied warranty of merchantability and fitness for any general or particular use. TUV India will not be liable for any claims for indirect, incidental or consequential damages including but not limited to loss of revenue, loss of profits and similar claims. TUV India will also not be liable under any strict liability, product liability or negligence. The total liability, scientifically and legally proven, if any, of TUV India, in any case, shall not exceed the invoiced amount for the services provided and paid for.
10. For Pesticide Residue analysis, reported value at LOQ level may vary within analytical variation of 50% considering Uncertainty of Measurement.

-- End of Report --



Manufacturing Site: Indore

Advanced Enzyme Technologies Ltd.

CIN: L24200MH1989PLC051018

Type 'B', Plot No. 5-13, Phase-1, Sector III, SEZ Indore,
Pithampur, District Dhar - 454 774, Madhya Pradesh, India.

Tel. / Fax: +91-7292-256 305

Email: info@advancedenzymes.com, Web.: www.advancedenzymes.com

Manufacturing Site: Nashik

Advanced Enzyme Technologies Ltd.

CIN: L24200MH1989PLC051018

Plot No. A-61/62, MIDC, Malegaon, Tal. Sinnar,
District Nashik - 422 113, Maharashtra, India.

Tel.: +91-997 010 0750 / +91-2551-230 044, Fax: +91-2551-230 816

Email: info@advancedenzymes.com, Web.: www.advancedenzymes.com

ALLERGEN INFORMATION

Product Name: ALTERNANSUCRASE

1. Cereals containing gluten, namely: wheat, rye, barley, oats, spelt, kamut or their hybridised strains, and products thereof, except⁽¹⁾: (a) wheat based glucose syrups including dextrose ; (b) wheat based maltodextrins; (c) glucose syrups based on barley; (d) cereals used for making alcoholic distillates including ethyl alcohol of agricultural origin;	NO ²
2. Crustaceans and products thereof;⁽¹⁾	NO
3. Eggs and products thereof;⁽¹⁾	NO
4. Fish and products thereof, except⁽¹⁾ (a) fish gelatine used as carrier for vitamin or carotenoid preparations; (b) fish gelatine or Isinglass used as fining agent in beer and wine	NO
5. Peanuts and products thereof;⁽¹⁾	NO
6. Soybeans and products thereof, except⁽¹⁾ (a) fully refined soybean oil and fat; (b) natural mixed tocopherols (E306), natural D-alpha tocopherol, natural D-alpha tocopherol acetate, and natural D-alpha tocopherol succinate from soybean sources; (c) vegetable oils derived phytosterols and phytosterol esters from soybean sources; (d) plant stanol ester produced from vegetable oil sterols from soybean sources;	NO ²
7. Milk and products thereof (including lactose), except⁽¹⁾ (a) whey used for making alcoholic distillates including ethyl alcohol of agricultural origin; (b) lactitol;	NO ²
8. Nuts⁽¹⁾, namely: almonds (<i>Amygdalus communis</i> L.), hazelnuts (<i>Corylus avellana</i>), walnuts (<i>Juglans regia</i>), cashews (<i>Anacardium occidentale</i>), pecan nuts (<i>Carya illinoensis</i> (Wangenh.) K. Koch), Brazil nuts (<i>Bertholletia excelsa</i>), pistachio nuts (<i>Pistacia vera</i>), macadamia or Queensland nuts (<i>Macadamia ternifolia</i>), and products thereof, except for nuts used for making alcoholic distillates including ethyl alcohol of agricultural origin;	NO
9. Sesame seeds and products thereof;⁽¹⁾	NO
10. Celery and products thereof;	NO
11. Mustard and products thereof;	NO
12. Sulphur dioxide and sulphites at concentrations of more than 10 mg/kg or 10 mg/litre in terms of the total SO ₂ which are to be calculated for products as proposed ready for consumption or as reconstituted according to the instructions of the manufacturers;	NO
13. Lupin and products thereof;	NO
14. Molluscs and products thereof.	NO

1. USA, top 9 of the Food Allergy Safety, Treatment, Education, and Research (FASTER) Act, 2021.

2. Wheat products (Wheat bran, Wheat flour) Soy products (Soya flour, soy grit and soya peptone) and milk products (Casein and Lactose) are used in the manufacturing facility in which this product is made.



SR. MANAGER QUALITY

Advanced Enzyme Technologies Ltd

Date: 13/04/2023

**Report of the GRAS Panel on the Generally Recognized as Safe (GRAS)
Status of the intended uses of the *Leuconostoc mesenteroides* enzyme
Alternansucrase expressed by *Escherichia coli* EBASSC**

September 2023

GRAS Panel Members

I. Glenn Sipes, Ph.D.
Michael W. Pariza, Ph.D.

Advisor to the GRAS Panel

Kevin O. Gillies

Introduction

Advanced Enzyme Technologies, LTD. (the company) proposes to use *Leuconostoc mesenteroides* alternansucrase, expressed by *Escherichia coli* EBASSC, to produce alternan oligosaccharides by enzymatic processes employing well known carbohydrate substrates. The company convened a panel of independent scientists (the GRAS panel), qualified by their scientific training and national and international experience to evaluate the safety of food ingredient, to conduct an independent and critical evaluation of the available information on its alternansucrase enzyme preparation, including its genetically modified production organism *E. coli* EBASSC and to determine whether the proposed uses of the enzyme preparation are safe and suitable and Generally Recognized as Safe (GRAS) based upon scientific procedures. The members of the GRAS Panel were Professors Michael W. Paria and I. Glenn Sipes, with Kevin O. Gillies serving as advisor to the GRAS Panel.

Summary and Basis for GRAS

The GRAS Panel, individually and collectively, critically evaluated data in a detailed dossier prepared by Advanced Enzymes, Ltd. entitled "Alternansucrase from *Leuconostoc mesenteroides* produced by *Escherichia coli* EBASSC" as updated in August 2023. This notice described (1) the identity and characterization of the enzyme; (2) the biology of *E. coli* K12 and its derivatives in the production of food ingredients, including non-pathogenicity, non-toxicogenicity, absence of antibiotic resistance genes, and acceptance of its use in food and industrial processes; (3) the well-characterized production strain *E. Coli* K12-derived *E. coli* EBASSC; (4) the alternansucrase gene from *Leuconostoc mesenteroides* and its protein product (alternansucrase); (5) the biology and safe history of use in food of the alternansucrase enzyme gene donor organism *Leuconostoc mesenteroides*; (6) the characteristics of the enzyme preparation, including stability, activity, usage rates, absence of the production organism and the absence of antimicrobial compounds in the enzyme preparation; (7) the manufacturing process; (8) food-grade product specifications; (9) data and use information indicating an absence of allergenicity concern; (10) toxicology testing and resulting No Observable Adverse Effect Level; (11) a highly conservative projected consumer intake of the enzyme preparation; and (12) formal evaluation by the Pariza-Johnson decision tree (MW Pariza and EA Johnson. Evaluating the Safety of Microbial Enzyme Preparations Used in Food Processing: Update for a New Century, *Regulatory Toxicology and Pharmacology* 33: 173-186, 2001), the conclusion of which indicated that the test article (alternansucrase enzyme preparation) was ACCEPTED.

The GRAS Panel participated in a teleconference on August 28, 2023 with Mr. Gillies, serving as technical advisor to the GRAS Panel. Following its independent and collective critical evaluation of the publicly available information, the GRAS Pael unanimously concluded that the proposed use of Advanced Enzyme Technologies Ltd.'s alternansucrase preparation as a processing aid based on scientific procedures is GRAS.

Conclusion

We, the GRAS Panel, independently and collectively critically evaluated the information and data presented in a dossier prepared by Advanced Enzymes Technologies Ltd. and summarized above and unanimously conclude that Advanced Enzyme Technologies Ltd.'s alternansucrase from *Leuconostoc mesenteroides* expressed by *Escherichia coli* EBASSC, manufactured in a manner that is consistent with current Good Manufacturing Practice for the production of food and meeting appropriate food-grade specifications, is safe and suitable and Generally Recognized as Safe (GRAS) based on scientific procedures for use as a processing aid in food production processes.

Further, it is our opinion that other experts qualified to assess the safety of food ingredients would concur with these conclusions.

Sep 1, 2023

Michael W. Pariza, Ph.D.
Emeritus Professor, Food Science
University of Wisconsin-Madison
Madison, Wisconsin

Date

Sep 1, 2023

I. Glenn Sipes, Ph.D. Fellow ATS and AAAS
Professor Emeritus, Pharmacology
University of Arizona
Tucson, Arizona

Date

Sep 1, 2023

Kevin O. Gillies (Advisor to the GRAS Panel)
Kevin O. Gillies Consulting Services, LLC
Denver, Colorado

Date

DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved:OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER 001250	DATE OF RECEIPT Jan 3, 2025
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3 Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): _____	
4 For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Ankit Kishor Rathi		Position or Title Sr. Manager - Regulatory Affairs	
	Organization (<i>if applicable</i>) Advanced Enzyme Technologies Ltd			
	Mailing Address (<i>number and street</i>) 5th Floor, A Wing Sun Magetica LIC Service Road, Louiiswadi			
City Thane (W)		State or Province Louiiswadi		Zip Code/Postal Code 400604
Country India				
Telephone Number +91 22 25830284		Fax Number		E-Mail Address ankit.rathi@advancedenzymes.com
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person KEVIN GILLIES		Position or Title MEMBER	
	Organization (<i>if applicable</i>) KEVIN O. GILLIES CONSULTING SERVICES, LLC			
	Mailing Address (<i>number and street</i>) 1759 GRAPE ST.			
City Denver		State or Province CO		Zip Code/Postal Code 80220
Country US				
Telephone Number 001-816-590-9836		Fax Number		E-Mail Address kevin@kogilliesconsulting.com

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Alternansucrase from *Leuconostoc mesenteroides* produced by *Escherichia coli* EBASSC

2. Submission Format: *(Check appropriate box(es))*

- ☒ Electronic Submission Gateway ☐ Electronic files on physical media
☐ Paper
If applicable give number and type of physical media _____

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

- ☐ Yes *(Proceed to Item 5)* ☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

- ☐ a) GRAS Notice No. GRN _____
☐ b) GRAS Affirmation Petition No. GRP _____
☐ c) Food Additive Petition No. FAP _____
☐ d) Food Master File No. FMF _____
☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

- ☒ Scientific procedures *(21 CFR 170.30(a) and (b))* ☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? *(see 21 CFR 170.225(c)(8))*

- ☐ Yes *(Proceed to Item 8)*
☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

- ☐ Yes, information is designated at the place where it occurs in the submission
☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

- ☐ Yes, a redacted copy of the complete submission
☐ Yes, a redacted copy of part(s) of the submission
☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

Alternansucrase from *E. coli* EBASSC is intended to be used in production of maltose alternan oligosaccharides, bakery process (bread making) and fruit processing for juice production.

The enzyme alternansucrase (E.C. 2.4.1.140) is intended to be used as a processing aid during food manufacturing. The enzyme is present in the food in insignificant amounts and does not have a technical effect in the final food. The enzyme preparation is used at the minimum levels necessary to achieve the desired effect and according to requirements under current Good Manufacturing Practices. The general population is the target population for consumption.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

- ☐ Yes ☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

- ☐ Yes ☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Advanced Enzyme Technologies Ltd.

(name of notifier)

has concluded that the intended use(s) of Alternansucrase from Leuconostoc mesenteroides produced by Esherichia coli EBASSC

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Advanced Enzyme Technologies Ltd. agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

5th Floor, 'A' wing, Sun Magnetica LIC Service Road, Louiswadi

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Kevin O. Gillies

Digitally signed by Kevin O. Gillies
Date: 2025.01.03 09:32:06 -07'00'

Printed Name and Title

Kevin Gillies, Consultant

Date (mm/dd/yyyy)

01/03/2025

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	AnnexA_Heavymetal_AISu.pdf	Administrative
	AnnexB_Allergenstatement_Alternansucrase.pdf	Administrative
	AnnexC_ReportoftheGRASPanelontheGenerallyRecognizedasSafe_Final_09012023-signed.pdf	Administrative
	GRASNotice_Alternansucrase_03Jan2025_Final.pdf	Administrative

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRAStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.